








Modelling Financial Markets

Chapter 17: Bubbles and Crashes



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Can Statistics Spot a Bubble Before It Bursts?

Bubbles, Crashes and Their Detection

- Prices sometimes rise far faster than any plausible change in fundamentals, then lose most of their value
- Questions
  - can statistics detect such episodes **in real time**?
  - can it say **when** they end?
- Part I: what a bubble is, run-ups and crashes in the data, tests for explosive prices and their date-stamping
- Part II: the LPPLS model (Log-Periodic Power Law Singularity), regime switching, crypto and artificial-intelligence (AI) stocks, a case study, one pass of the AI-assisted research loop
- Technical detail sits in the Appendix at the end

What you should be able to do

- separate the fundamental value from a rational bubble and state what can be tested
- test for explosive prices on rolling windows and date-stamp the episodes, with valid critical values
- fit LPPLS, judge the stability of its critical time and evaluate its alarms honestly
- describe booms and busts with a Markov-switching model and test the number of regimes

Core Path, Applications and Proofs

- **Core path**: the main slides of four sections, in order
  - What Is a Bubble?, Testing for Explosive Behaviour, the LPPLS model, Regime-Switching Models; then Which Tool When?
  - Two hand-worked examples carry the core logic: why the test uses log prices, and how a run of exceedances becomes a dated episode
- **Applications**, read after the core
  - Run-ups and crashes in the data, crypto and AI stocks, the housing case study, the AI-assisted research loop
- ▸ Appendix, optional depth
  - Derivations, limit theory, estimation details and caveats; the blue buttons on the main slides lead there and back
- **Seminar 17**: numbers cited next to an exercise are replication targets
  - Reproduce the number first; if yours differs, compare the sample period, the minimum window and the critical values before the code

Three Historical Lessons



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain; Wikimedia Commons

- **Tulips, 1636–1637** Garber (1990)
  - Contracts on rare bulbs traded at very high prices until the market stopped in February 1637
  - Part of the path fits the fundamentals of a new luxury good
  - Lesson 1: calling something a bubble requires a model of the fundamental value
- **South Sea and Mississippi, 1720** Kindleberger & Aliber (2005)
  - Shares rose several-fold within months, then collapsed
  - Ingredients: a financial innovation, easy credit, a story about future profits
  - Lesson 2: bubbles come with credit and with high turnover Scheinkman & Xiong (2003)
- **1929** Kindleberger & Aliber (2005)
  - A boom partly financed with borrowed money ended in the October crash
  - Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic
  - Lesson 3: we need tests that work on a single time series, in real time, before the panic

The Series Tested: Stocks, Industries, Crypto

Series	Frequency	Period	Use
S&P Composite, real price and dividend	monthly	1871–2026	price-dividend ratio
48 US industry portfolios	monthly	1926–2026	run-ups and crashes
Nasdaq 100, S&P 500	daily, monthly	1990–2026	dot-com, AI
BET, BET-FI	daily, monthly	1997–2026	Bucharest Stock Exchange indices
Bitcoin, Ethereum, Dogecoin	daily, weekly	2014–2026	crypto booms
NVIDIA, Cisco, Tesla, GameStop, Strategy (formerly MicroStrategy)	daily, monthly	1990–2026	single stocks

- Closing values up to 18 September 2026; stocks adjusted for dividends and splits; weekly and monthly series use the last price of the period
- Sources: shillerdata.com for the S&P Composite; Kenneth French Data Library for industries

Fundamental Value

- No-arbitrage pricing: $P_t = \mathbb{E}_t[M_{t+\mathbf{1}}(P_{t+\mathbf{1}} + D_{t+\mathbf{1}})]$
  - P_t: price at the end of period t; $D_{t+\mathbf{1}}$: dividend paid in period $t+1$
  - $M_{t+\mathbf{1}}$: stochastic discount factor, the random factor that converts a payoff at $t+1$ into its value at t; $\mathbb{E}_t$: expectation under the physical measure (the probabilities of actual outcomes), given the information at t
- **Model assumption**: a constant required (expected) return r
  - Then $P_t = \mathbb{E}_t[P_{t+\mathbf{1}} + D_{t+\mathbf{1}}]/(1+r)$: today's price is the discounted expected price plus dividend
  - In effect $M_{t+\mathbf{1}}$ is replaced by the constant $1/(1+r)$; the rest of the chapter uses only this case
- Solving forward gives the **fundamental value**: $F_t = \sum_{i \geq \mathbf{1}} \mathbb{E}_t[D_{t+i}]/(1+r)^i$
  - The present value of all expected dividends
  - Gordon case, dividends growing at $g < r$: $F_t = D_{t+\mathbf{1}}/(r-g)$
- But $P_t = F_t + B_t$ also solves the pricing equation
  - B_t: any extra term with $\mathbb{E}_t[B_{t+\mathbf{1}}] = (1+r)\,B_t$
  - B_t is the **rational bubble**: investors pay for it because they expect to resell at a higher price Blanchard & Watson (1982)

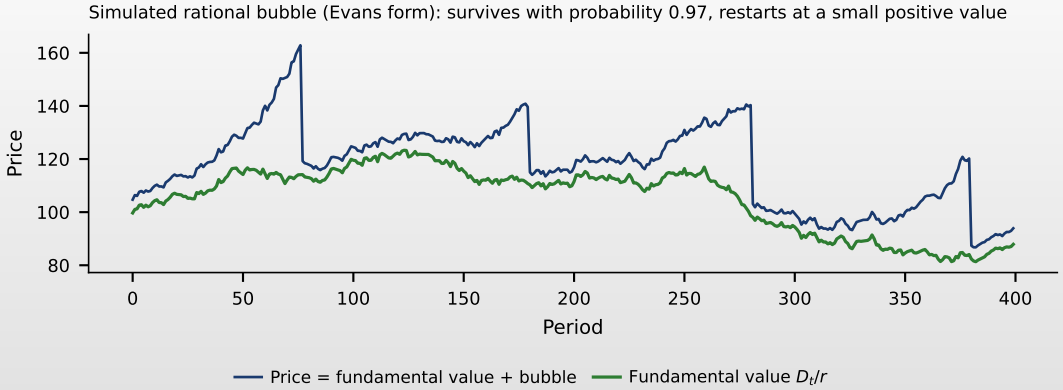
Why a Rational Bubble Must Explode

- From $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$: $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$ for any horizon of h periods, an **explosive conditional mean**; individual paths can crash
  - The bubble grows faster than dividends ($r > g$), so its share of the price rises over time
- Rational bubbles are positive and exist from the first trade Diba & Grossman (1988)
  - A negative bubble would eventually push the expected price below zero
  - A bubble that is zero today must stay zero
- A bubble is the failure of the **transversality condition** $\lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$
  - The price contains a value never paid out as dividends (derivation in the Appendix)
- **Testable implication**
  - If dividends are at most I(1), then without a bubble the price and the price-dividend ratio are **not explosive**
  - I(1): integrated of order one; the first difference is I(0) (stationary), while the level is not

Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability π and collapses with probability $1-\pi$ Blanchard & Watson (1982)
  - $B_{t+1} = \dfrac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π, $B_{t+1} = \varepsilon_{t+1}$ otherwise
  - $\pi \in (0,1)$: survival probability; ε_{t+1}: a shock with mean zero given the information at t
- Expected growth is still r; while it survives the bubble grows faster, at $(1+r)/\pi - 1$, to pay for the crash risk
- Expected lifetime: $1/(1-\pi)$ periods, a geometric waiting time
- After a collapse $B = \varepsilon$ can be negative, which Diba–Grossman rule out: the simulation on the next slide restarts from a positive value (Evans form) Evans (1991)

A Simulated Rational Bubble (1/2)



MFM_ch17_rational_bubble

A Simulated Rational Bubble (2/2)

- Fundamental value D_t/r, random-walk dividends; $r = 1\%$, $\pi = 0.97$
  - While alive and large the bubble grows by about 4.1% per period; expected life 33.3 periods
- Evans form: after a collapse the bubble restarts from $\delta = 5$ Evans (1991)
  - Survival (probability π): $B_{t+\mathbf{1}} = \left[\delta + \frac{\mathbf{1}+r}{\pi}(B_t - \frac{\delta}{\mathbf{1}+r})\right] u_{t+\mathbf{1}}$; collapse (probability $1-\pi$): $B_{t+\mathbf{1}} = \delta\, u_{t+\mathbf{1}}$
  - $u_{t+\mathbf{1}}$: noise with mean 1, drawn independently of survival; hence $\mathbb{E}_t[B_{t+\mathbf{1}}] = \pi\left[\delta + \frac{\mathbf{1}+r}{\pi} B_t - \frac{\delta}{\pi}\right] + (1-\pi)\delta = (1+r)B_t$
  - The simulation uses these two branches at every B_t, without the small-bubble threshold of Evans' original form
- In 400 periods the bubble grew large and burst 4 times; at its largest the price was 1.43 times the fundamental value
- Between episodes the price tracks fundamentals
  - A full-sample test sees mostly "normal" data: the tests of this chapter use **windows**

MFM_ch17_rational_bubble

Bubbles Beyond Rational Expectations



Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017); CC BY 2.0; Wikimedia Commons

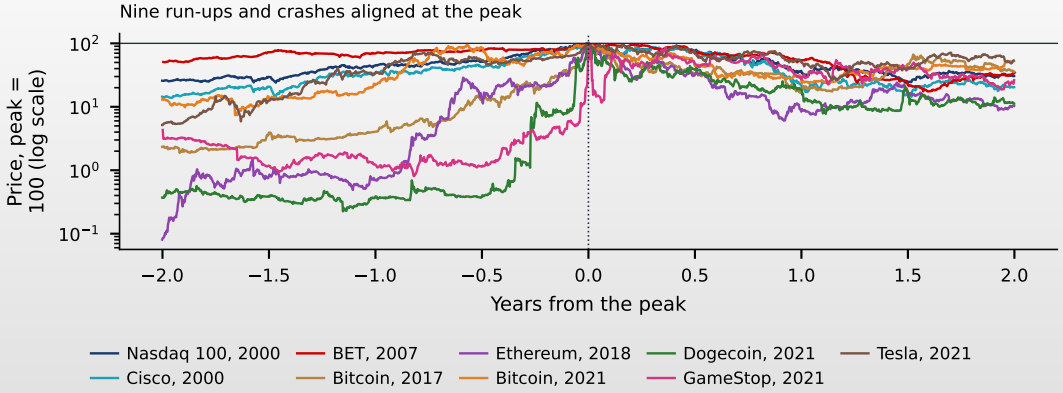
- **Feedback and narratives** Shiller (2015)
  - Rising prices attract new buyers
  - "New era" stories spread by word of mouth
- **Disagreement and short-sale limits** Scheinkman & Xiong (2003)
  - The price includes the option to resell to a more optimistic buyer
- **Riding the bubble** Abreu & Brunnermeier (2003), Brunnermeier & Nagel (2004)
  - Informed traders stay in as long as others do, since no one can burst it alone
- Common signature of all three
  - Prices grow faster than fundamentals for a while: the explosive pattern we will test

Nine Episodes: Run-Up and Crash

Episode	Peak	2-year run-up	Max. fall	Trough	Back at peak
Nasdaq 100, 2000	27 Mar 2000	+288%	−83%	Oct 2002	Nov 2015
Cisco, 2000	27 Mar 2000	+592%	−89%	Oct 2002	Aug 2021
BET, 2007	24 Jul 2007	+97%	−83%	Feb 2009	Mar 2021
Bitcoin, 2017	16 Dec 2017	+4 186%	−83%	Dec 2018	Nov 2020
Ethereum, 2018	13 Jan 2018	+124 004%	−94%	Dec 2018	Feb 2021
Bitcoin, 2021	8 Nov 2021	+667%	−77%	Nov 2022	Mar 2024
Dogecoin, 2021	7 May 2021	+26 897%	−92%	Jun 2022	not yet
GameStop, 2021	27 Jan 2021	+2 218%	−88%	Apr 2024	not yet
Tesla, 2021	4 Nov 2021	+1 837%	−74%	Jan 2023	Dec 2024

- Peak: highest daily close in the episode; run-up: simple percentage change of the price over the two years before the peak, $P_{\mathbf{peak}}/P_{\mathbf{peak}-\mathbf{2y}} - 1$; maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years, yet with one of the deepest falls

Run-Ups and Crashes Aligned at the Peak (1/2)



MFM_ch17_episodes

Run-Ups and Crashes Aligned at the Peak (2/2)

- Daily prices two years before and after each peak, peak = 100, log scale
- Two-year run-ups differ enormously, from +97% (BET) to +124 004% (Ethereum); the falls are much more alike, 74–94%
- Common shape: an accelerating rise, a sharp top, a deep fall
  - Convex on the log scale: the growth rate itself rises, the **explosive** behaviour the tests look for
- The run-ups in the table are simple percentage changes; the tests use log returns $\ln(P_t/P_{t-1})$
  - Example: a doubling is +100% as a simple change, but $\ln 2 \approx 0.69$ as a log return
  - Log returns add up over time and match the log scale of the chart; simple changes compound
- Selection bias: we picked these episodes because they crashed
  - Many run-ups were not followed by crashes

MFM_ch17_episodes

The Dot-Com Bubble



The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

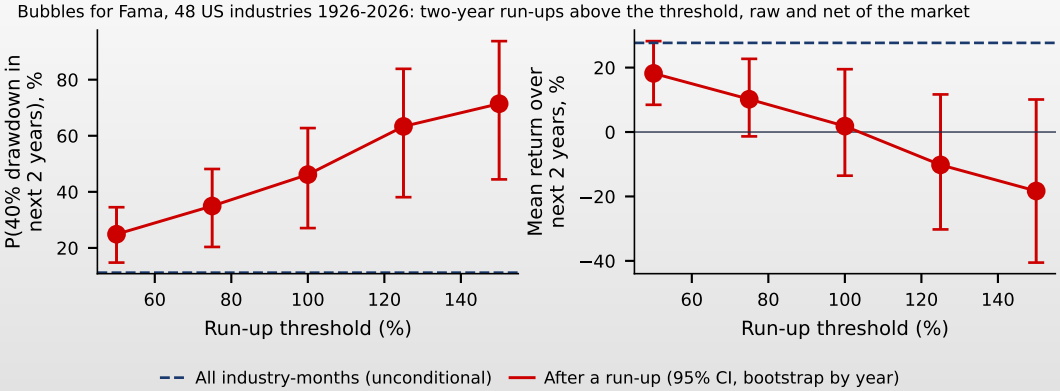
Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia Commons

- After the peak of 27 Mar 2000: Nasdaq 100 −83%, Cisco −89% (table)
- Hedge funds rode the bubble and cut positions before the fall Brunnermeier & Nagel (2004)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble Pástor & Veronesi (2009)

Bubbles for Fama: The Design

- Fama's objection: a "bubble" is only a bubble if its end is predictable; otherwise it may be a rational reaction to news Fama (2014)
- Greenwood, Shleifer and You answer with **all** US industries since 1926, not with famous episodes only: do sharp run-ups predict crashes, or low returns? Greenwood, Shleifer & You (2019)
- **Run-up**: value-weighted industry return (each firm weighted by its market capitalisation) over the past 2 years above a threshold (e.g. 100%)
  - Both conditions at once: the raw return **and** the return net of the market exceed the threshold
  - And a raw return of at least 50% over the past 5 years (from 1931), so that recoveries from slumps are not counted (Section 2 of the paper)
  - First month of the run-up; a new episode in the same industry only two years later
- **Crash**
  - A drawdown of at least 40% from a previous high within the next 2 years
  - The next chart gives each crash frequency with a 95% CI (Confidence Interval)
- Our replication, same sample rules: the first 48 of the 49 industries of the Kenneth French Data Library (without "Other"), months with at least ten firms: 41 679 industry-months, 1926–2026

What Do Sharp Run-Ups Predict? (1/2)



MFM_ch17_bubbles_for_fama

What Do Sharp Run-Ups Predict? (2/2)

- After a run-up above 100% (52 episodes in 27 industries), a 40% crash followed in 46% of cases, 95% CI [27%, 63%]
  - Run-up above 100% both raw and net of the market; the five chart points are the thresholds 50, 75, 100, 125 and 150%
  - Against 11% for a random industry-month; above 150% (rightmost point): 71% ($n = 21$)
  - CI: bootstrap over calendar years, because run-ups cluster in time (e.g. 1928–1929, 1967–1968, 1980); effective sample size in the Appendix
- The mean return over the next 2 years is not significantly different from zero
  - 2% after a 100% run-up, CI $[-14\%, 20\%]$: no statistically established mean loss
- Run-ups raise the **risk** of a crash, not the predictable loss: price run-ups alone do not give a profitable timing signal Greenwood, Shleifer & You (2019)

MFM_ch17_bubbles_for_fama

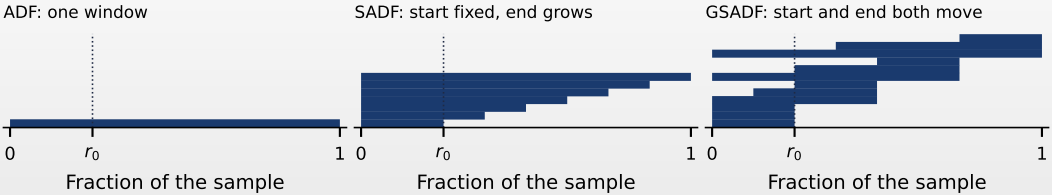
Right-Tailed ADF and Collapsing Bubbles

- Regression on the log price y_t: $\Delta y_t = a + b\, y_{t-1} + \varepsilon_t$
  - $\Delta y_t = y_t - y_{t-1} = \ln(P_t/P_{t-1})$: the log return; a: constant; ε_t: shock; here no lagged Δy
  - $\mathrm{ADF} = \hat{b}/\mathrm{SE}(\hat{b})$: the Augmented Dickey–Fuller (ADF) t-statistic, called t_b in Seminar 17; $\hat{b}$: the OLS (Ordinary Least Squares) estimate; SE: its Standard Error
  - $b < 0$: mean reversion; $b = 0$: random walk (unit root); $b > 0$: each rise feeds the next, an explosive path
- Bubble test: $H_0 : b = 0$ (unit root) vs $H_1 : b > 0$ (explosive root), the **right tail** Phillips, Wu & Yu (2011)
  - The left-tail Dickey–Fuller table does not apply
- Evans: a bubble that collapses periodically leaves the **full-sample** test with low power Evans (1991)
  - The whole series looks stationary or like a random walk
  - Collapses give the full-sample slope a negative sign (Appendix)
- Solution: run the ADF regression on many **windows** and take the largest statistic
  - SADF (Sup Augmented Dickey–Fuller) Phillips, Wu & Yu (2011) and GSADF (Generalised Sup Augmented Dickey–Fuller) Phillips, Shi & Yu (2015a) differ in the windows they search (next two slides)

Levels Explode, Log Prices Accelerate

- Theory: the bubble is explosive in **levels**, $\mathbb{E}_t[B_{t+\mathbf{1}}] = (1+r)B_t$; the test regresses the **log** price
  - A pure bubble path $B_t = B_{\mathbf{0}}(1+r)^t$ has a constant log growth $\ln(1+r)$: Δy_t does not depend on $y_{t-\mathbf{1}}$, so $b = 0$
  - What $b > 0$ detects: a **rising** growth rate of $\ln P_t$, which occurs while the bubble share of $P_t = F_t + B_t$ grows
- Toy example ($1 + r = 2$ for easy arithmetic): $F = 100$ constant, $B = 1, 2, 4, 8$, so $P = 101, 102, 104, 108$
  - Bubble share of the price: 1.0%, 2.0%, 3.8%, 7.4%
  - Log increments: $\ln(102/101) = 0.0099$, $\ln(104/102) = 0.0194$, $\ln(108/104) = 0.0377$
  - They rise with $y_{t-\mathbf{1}} = \ln P_{t-\mathbf{1}}$, so the fitted $b > 0$; the bubble alone, $B_t = 2^t$, has every increment equal to $\ln 2 = 0.693$, so $b = 0$
- Check: add $B = 16$, so $P = 116$; is the next increment larger?
  - Yes, $\ln(116/108) = 0.0715$; once $B \gg F$ the increments approach $\ln 2$ and b falls back towards 0: windows that catch the **transition** have the power
- Derivation and the role of the deterministic terms: Explosive in Levels, Accelerating in Logs

Three Ways to Choose Windows (1/2)



- $\mathrm{ADF}_{r_1}^{r_2}$: the ADF statistic on the window from fraction r_1 to fraction r_2 of the sample
  - Each bar in the chart is one ADF regression
  - r_0: the minimum window, as a fraction of the T changes Δy_t (observations minus one); sup: the largest value over the windows allowed
- Minimum window rule: $r_0 = 0.01 + 1.8/\sqrt{T}$ Phillips, Shi & Yu (2015a)
  - E.g. $T = 440$ months: $r_0 = 0.096$, i.e. 42 months

MFM_ch17_windows

Three Ways to Choose Windows (2/2)

- SADF Phillips, Wu & Yu (2011)
  - $\mathrm{SADF}(r_0) = \sup_{r_2 \in [r_0, 1]} \mathrm{ADF}_0^{r_2}$
  - Windows start at the first observation and end anywhere after r_0
- GSADF Phillips, Shi & Yu (2015a)
  - $\mathrm{GSADF}(r_0) = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$
  - Windows start and end anywhere, with length at least r_0
- BSADF (Backward Sup Augmented Dickey–Fuller): backward SADF at the end point r_2 Phillips, Shi & Yu (2015a)
  - $\mathrm{BSADF}_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} \mathrm{ADF}_{r_1}^{r_2}$, $\mathrm{GSADF} = \sup_{r_2} \mathrm{BSADF}_{r_2}$
  - BSADF_{r_2} uses only data up to r_2: it can be computed **in real time**
- Larger search, higher bar
  - The more windows a statistic searches, the larger its critical value: simulated for each T and r_0

Date-Stamping an Episode (1/2): the PSY Rules

- **Origination**, in the PSY (Phillips, Shi and Yu) procedure Phillips, Shi & Yu (2015a)
  - The first date at which BSADF_{r_2} exceeds its critical value cv_{r_2}
  - cv_{r_2}: the 95% quantile of BSADF_{r_2} under the null, date by date
- **Termination** (PSY, eq. 8) Phillips, Shi & Yu (2015a)
  - The first date after $\hat{r}_e + \delta \ln(T)/T$ at which BSADF falls below cv_{r_2}
  - $\hat{r}_e$: the estimated start, as a fraction of the sample; T: number of changes Δy_t
  - $\delta \ln(T)/T$: a minimum duration, with δ chosen for the data frequency
- PWY (Phillips, Wu and Yu) date-stamping uses the recursive $\mathrm{ADF}_0^{r_2}$ instead Phillips, Shi & Yu (2015b)
  - It may miss a second bubble after a collapse

Date-Stamping an Episode (2/2): the Rule Used Here

- Minimum-duration rule of the R package exuber Vasilopoulos, Pavlidis & Martínez-García (2022)
  - Runs of consecutive exceedances shorter than L periods are dropped
  - $L = \lceil \ln T \rceil$, i.e. $\delta = 1$; $\lceil \cdot \rceil$: rounded up; $T = 440$ months gives $L = 7$
- Start and end of an episode
  - Start: the first exceedance; end: the last exceedance (exuber dates the end one period later)
  - Eq. 8 would instead extend an isolated one-date exceedance to L dates
- Real time
  - The start is dated in hindsight
  - The episode is **confirmed** only after L consecutive exceedances, $L - 1$ periods after its start
- **Chain**: kept runs separated by a short gap, used for the leads in Real Time vs Hindsight
  - Monthly data: the next run starts at most 93 days after the previous one ends (at most two months without an exceedance)

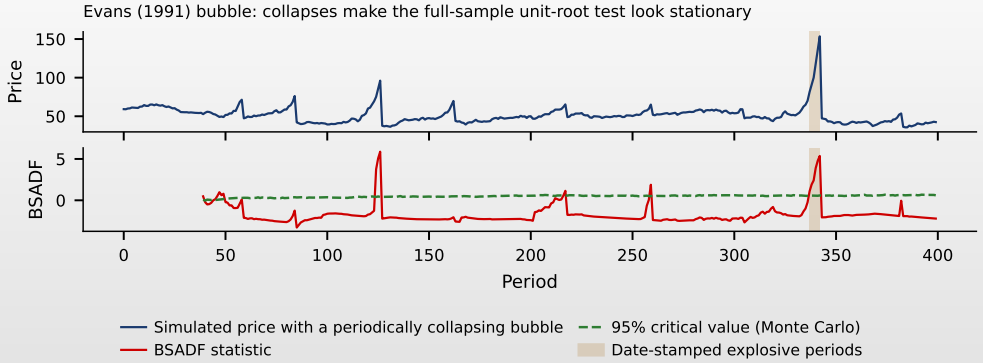
A Hand-Worked Date-Stamping Example

- Invented numbers; $L = 3$; critical value $cv_t = 1.0$ at every date; flag $= 1$ if $\mathrm{BSADF}_t > cv_t$; s: start, c: confirmed, e: end, d: dropped

t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
BSADF_t	0.4	1.2	1.5	1.8	1.3	0.8	1.1	1.4	1.9	1.6	0.7	0.5	1.3	1.2	0.6
Flag	0	1	1	1	1	0	1	1	1	1	0	0	1	1	0
Event		s		c	e		s		c	e			d	d	

- Runs $t = 2$–5 and $t = 7$–10: four exceedances each, at least L, so both are kept
  - Confirmed at $t = 2 + L - 1 = 4$ and $t = 7 + L - 1 = 9$; each ends at its last exceedance, $t = 5$ and $t = 10$
  - Run $t = 13$–14: only two exceedances, fewer than L: dropped, never confirmed
- A toy chain allowing at most one quiet date between runs links 2–5 and 7–10 (one quiet date, $t = 6$). The monthly rule allows up to two quiet months (93 days)
  - Episode containing $t = 10$: strict rule start $t = 7$; chain start $t = 2$, so the measured lead grows by five periods
- Check: flags 1 1 0 1 1 1 1 with $L = 3$; when is the first episode confirmed? Answer: at $t = 6$; the run at $t = 1$–2 is too short

Evans Bubble: Full Sample vs Windows (1/2)



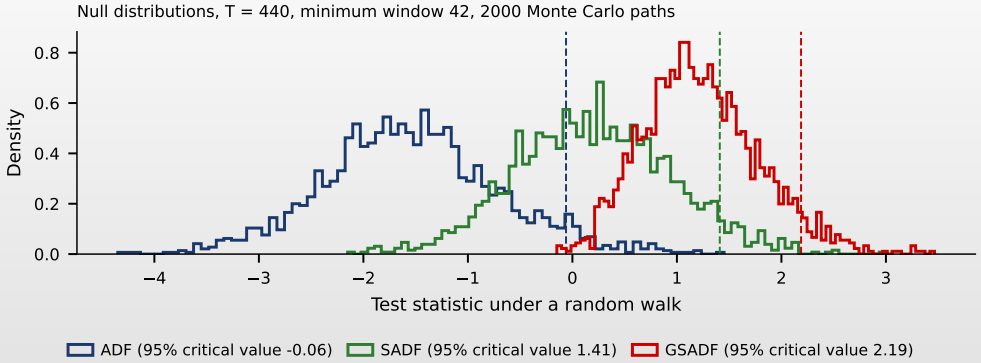
MFM_ch17_evans

Evans Bubble: Full Sample vs Windows (2/2)

- 400 simulated periods; BSADF against its 95% critical value from 2 000 random-walk paths
- Full-sample right-tailed ADF: -5.49 vs 95% critical value -0.09: no rejection, the bubble is invisible
- SADF 0.98 (critical value 1.46) and GSADF 5.88 (critical value 2.21)
- BSADF crosses its critical value in several bursts, but only one lasts long enough to be date-stamped
  - The minimum-duration rule ignores short bursts and so reduces false alarms

MFM_ch17_evans

Null Distributions and Critical Values (1/2)



MFM_ch17_windows

Null Distributions and Critical Values (2/2)

- Random walk with a small drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, 2 000 paths of $T = 440$; 95% critical values: a single ADF -0.06 (negative!), SADF 1.41, GSADF 2.19 Phillips, Shi & Yu (2015a)
- A supremum over more windows shifts the distribution to the right: the GSADF bar is the price of searching over about 80 000 windows; simulate it for each T, r_0 and lag order
- Asymptotics (▸ Appendix): the null limit is pivotal; against mildly explosive roots the tests are consistent; date-stamping keeps a 5% false-alarm chance at every date Phillips, Shi & Yu (2015b)

MFM_ch17_windows

Changing Volatility: Wild Bootstrap

- Monte Carlo critical values assume shocks with constant variance
  - Real prices show volatility clustering and shifts in volatility
  - A burst of volatility can mimic explosiveness and inflate the rejection rate Harvey et al. (2016)
- **Wild bootstrap** Phillips & Shi (2020)
  - Multiply the observed, demeaned changes Δy_t by random signs $w_t = \pm 1$, rebuild the path, recompute BSADF
  - The bootstrap data have a unit root by construction (no explosive root) and inherit the time pattern of $\mathrm{Var}(\Delta y_t)$ Harvey et al. (2016)
  - Demeaning removes the drift, as in the null model
- Retrospective and real-time versions
  - Centring with the full-sample mean and resampling the whole path give **retrospective** critical values
  - In real time, recompute them with data up to each date

Size of GSADF When Volatility Changes: Design

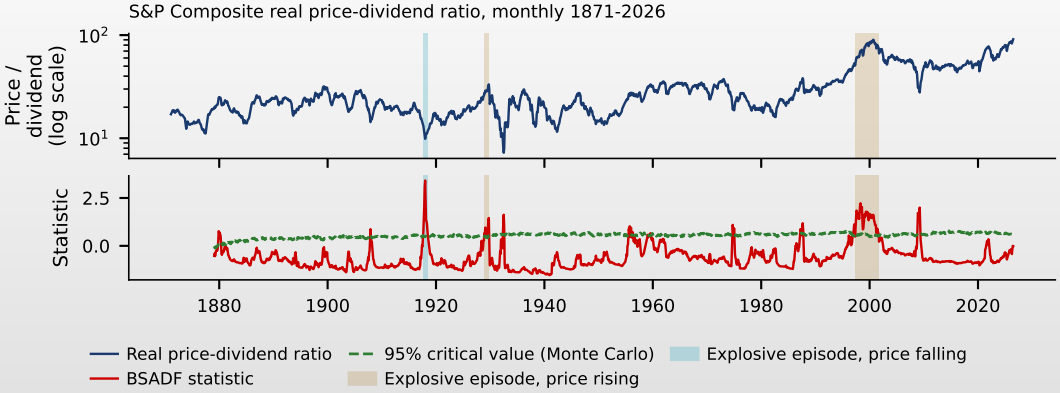
- Null model without a bubble: $y_t = T^{-1} + y_{t-1} + e_t$, a random walk with a small drift
  - e_t: shocks with mean zero and the variance pattern of each scenario
- Four variance patterns
  - Constant; multiplied by 3 at $T/2$; divided by 3 at $T/2$
  - GARCH (Generalised AutoRegressive Conditional Heteroskedasticity)(1,1): $\sigma_t^2 = \omega + \alpha e_{t-1}^2 + \beta \sigma_{t-1}^2$, with $\alpha = 0.10$, $\beta = 0.88$
  - σ_t^2: conditional variance; $\omega > 0$: constant; α: reaction to the last shock; β: persistence
- Sample: $T = 400$, minimum window $w_0 = \lfloor r_0 T \rfloor = 40$ observations
  - 1 000 paths per scenario, each with its own wild bootstrap (199 replications)
  - Monte Carlo error: ± 1.4 percentage points near 5%
- Two outcomes per path
  - Size: GSADF above its 95% critical value (Monte Carlo, MC in the table, or wild bootstrap)
  - False episode: BSADF above its pointwise 95% value for 6 consecutive periods

Size of GSADF When Volatility Changes: Results

Null model, no bubble	GSADF, MC crit. value	GSADF, wild crit. value	False episode, MC	False episode, wild
Constant volatility	4.1%	3.9%	49.1%	51.9%
Volatility $\times 3$ at $T/2$	32.0%	4.1%	63.8%	49.6%
Volatility $\times 1/3$ at $T/2$	4.3%	6.1%	48.0%	54.8%
GARCH(1,1), $\alpha = 0.10$, $\beta = 0.88$	16.6%	6.8%	58.1%	52.8%

- A late rise in volatility is mistaken for explosiveness Harvey et al. (2016)
  - Size 32.0% with Monte Carlo, 4.1% with wild bootstrap critical values
  - Under GARCH the wild bootstrap removes most, not all, of the distortion; a fall in volatility is harmless
- Even with constant volatility, 49.1% of the paths contain a false date-stamped episode
  - The size is controlled date by date, not over the whole sample
- The GARCH and $\times 1/3$ rows are the replication target of Seminar 17, B9

S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026 (1/2)



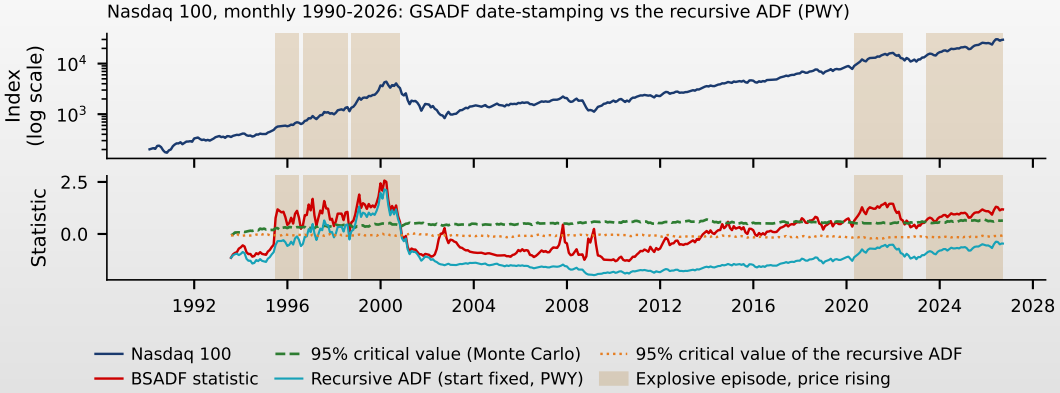
MFM_ch17_sp_pd

S&P Composite (Shiller) Price-Dividend Ratio, 1871–2026 (2/2)

- Real price over real dividend, monthly, Jan 1871–Jun 2026
  - $T = 1\,865$, minimum window 96 months
  - Shiller's S&P Composite, not the S&P 500 series; in the chart legend, "price rising/falling" refers to the plotted series, here the ratio
- GSADF $= 3.41$ vs 95% critical value 2.39 (99%: 2.88)
  - The full-sample ADF (-1.11) sees nothing
- Episodes: Aug 1917–Jun 1918; Nov 1928–Sep 1929; May 1997–Aug 2001
  - 1917–1918: an accelerating **fall** of the ratio, explosive but not a bubble
- Dot-com: peak 90.2 in Aug 2000, long-run mean 29.8
  - The sample maximum, 91.2 in May 2026, lies in no episode: a high ratio is not necessarily explosive
- A ratio can explode through its denominator: check both price and dividend

MFM_ch17_sp_pd

Nasdaq 100: GSADF vs PWY Date-Stamping (1/2)



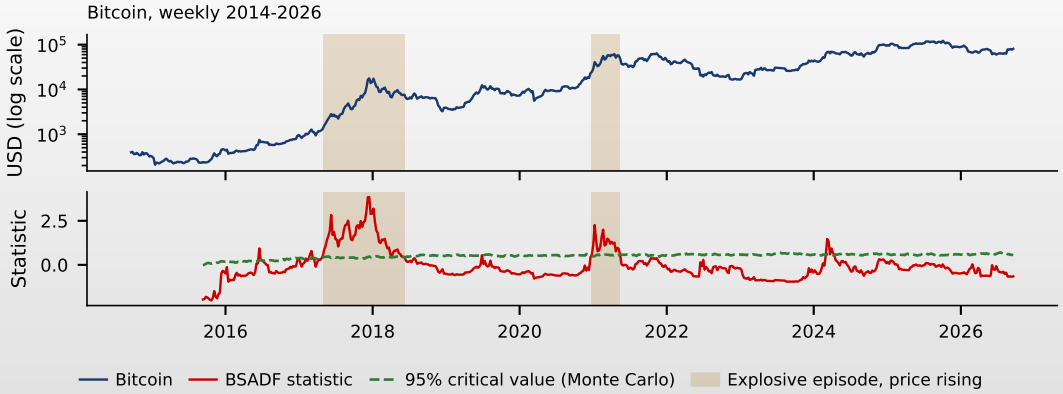
MFM_ch17_nasdaq

Nasdaq 100: GSADF vs PWY Date-Stamping (2/2)

- Monthly log closes 1990–2026, $T = 440$
  - Minimum window 42 months; 2 000 Monte Carlo paths
- GSADF $= 2.57$ (95% critical value 2.19); SADF $= 2.14$ (1.41)
- BSADF episodes: Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000; Apr 2020–May 2022; May 2023–ongoing
  - PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
- The recursive ADF misses the post-2020 episodes
  - It starts in 1990 and is dragged down by the long flat period after 2000
- Dot-com episode dated from Jun 1995: the start of a chain of 3 BSADF runs
  - 57 months before the peak of Mar 2000; confirmed in real time in Dec 1995
  - The run that contains the peak alone starts in Sep 1998

MFM_ch17_nasdaq

Bitcoin: Explosive Episodes (1/2)



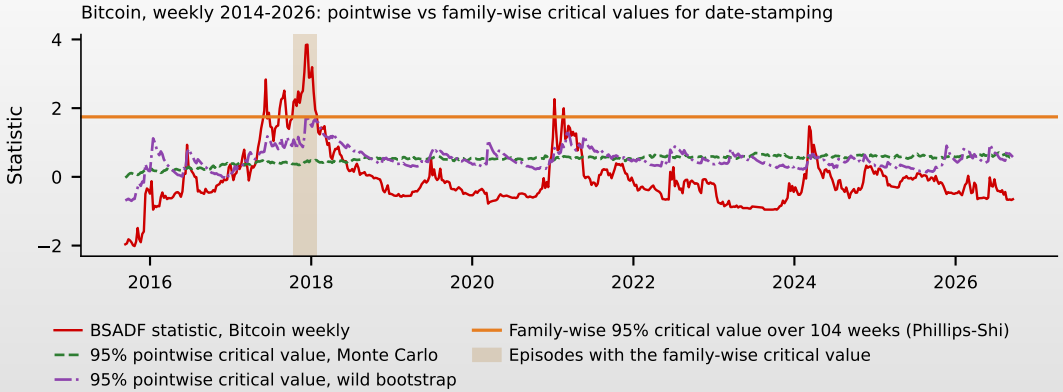
MFM_ch17_bitcoin

Bitcoin: Explosive Episodes (2/2)

- Weekly log closes in US dollars (USD) 2014–2026, $T = 626$
  - Minimum window 51 weeks; Monte Carlo critical values
- GSADF $= 3.85$: rejected with either critical value
  - 95% critical value: Monte Carlo 2.27, wild bootstrap 2.82
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
  - Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- 2017: episode dated from Apr 2017
  - 8 months before the peak of 16 Dec 2017; confirmed in real time in Jun 2017
  - It ended 6 months after the peak
- The 2024–2025 rise to new highs is **not** flagged
  - Steady growth is not explosive acceleration

MFM_ch17_bitcoin

Date-Stamping Is a Multiple Test: Bitcoin (1/2)



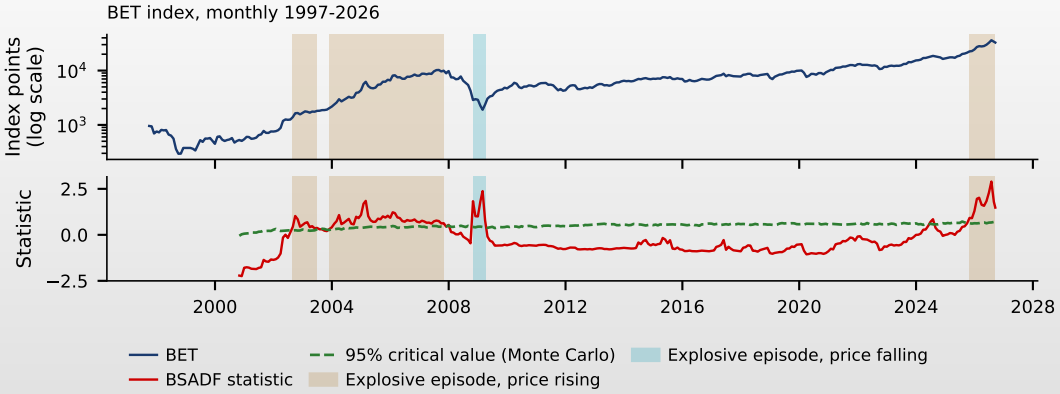
MFM_ch17_inference

Date-Stamping Is a Multiple Test: Bitcoin (2/2)

- BSADF vs its critical value at each of 626 weeks: hundreds of dependent tests
- Family-wise error rate (FWER)
  - The probability of at least one false alarm over the whole monitoring window
  - 2 000 random walks with the Bitcoin T and w_0: a crossing in 95%, an "episode" in 59% of paths
- Phillips and Shi: bootstrap the **maximum** of BSADF Phillips & Shi (2020)
  - Over the last τ_b monitoring dates; here $\tau_b = 104$ weeks
  - Family-wise critical value 1.75, against Monte Carlo pointwise values of at most 0.71; wild-bootstrap pointwise values can be higher because they retain time-varying volatility, unlike this i.i.d. residual bootstrap
- Episodes above the family-wise threshold: Oct 2017–Jan 2018
  - The 2020–2021 episode found with Monte Carlo pointwise values does not survive
- A sensitivity check, not proved FWER control (design in the Appendix)

MFM_ch17_inference

BET: The 2007 Boom and 2025 (1/2)



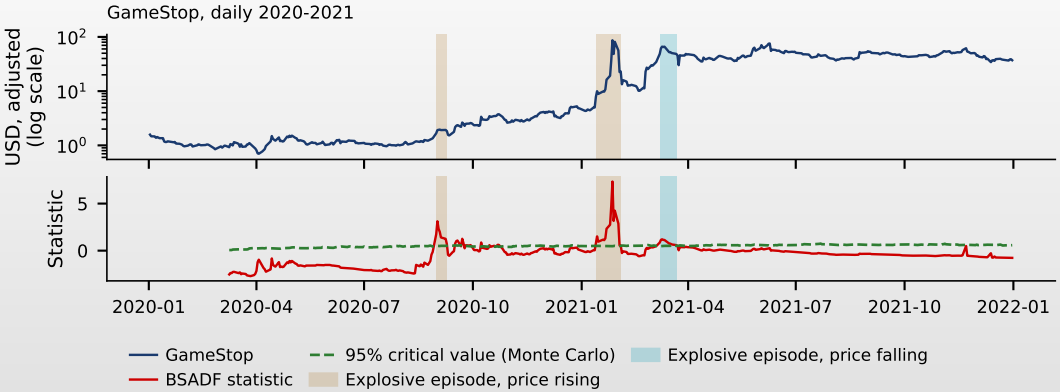
MFM_ch17_bet_gamestop

BET: The 2007 Boom and 2025 (2/2)

- Monthly log closes of the official BET, 1997–2026
  - $T = 348$, minimum window 37 months
- GSADF $= 2.90$ vs 95% critical value 2.21
  - Episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
- 2003–2007: the run-up around Romania's accession to the European Union (January 2007)
  - BET then fell 83% from its peak of 24 Jul 2007
- Oct 2008–Mar 2009: an accelerating **fall**
  - Explosive can mean down as well as up
- The current episode (from Oct 2025) is still open
  - Explosive does not mean it will crash: the rise is accelerating

MFM_ch17_bet_gamestop

GameStop, January 2021 (1/2)



MFM_ch17_bet_gamestop

GameStop, January 2021 (2/2)



A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

- Daily log closes 2020–2021, $T = 504$, window 45 days; GSADF $= 7.34$, critical value 2.21
- January episode dated from 13 Jan 2021, 14 days before the closing high of 27 Jan 2021; confirmed on 22 Jan 2021
- A surge of a few weeks: detecting it needs daily data
  - The test does not identify the mechanism (short covering: buying back shares sold short)

MFM_ch17_bet_gamestop

Real Time vs Hindsight

Episode	Start / Peak	Confirmed	Lead (months)	Lead from conf.	Rise after conf.	Fall after peak
Nasdaq 100, monthly	Jun 1995 / Mar 2000	Dec 1995	57	51	+716%	−76%
BET, monthly	Nov 2003 / Jul 2007	Apr 2004	44	39	+299%	−83%
Bitcoin 2017, weekly	Apr 2017 / Dec 2017	Jun 2017	8	6	+590%	−83%
Bitcoin 2021, weekly	Dec 2020 / Apr 2021	Feb 2021	4	2	+66%	−75%
GameStop, daily	13 Jan 2021 / 27 Jan 2021	22 Jan 2021	0.5	0.2	+435%	−88%

- Columns (caveats in the ▸ Appendix)
  - Start: first exceedance of the chain of BSADF runs that leads to the peak (hindsight; chain defined on the next slide); confirmed: after L consecutive exceedances (real time)
  - Leads in months; rise: from confirmation to the peak; fall: to the lowest close within 2 years
- Even at confirmation the tests **warn early**, but they do not time the crash Phillips, Shi & Yu (2015b)
  - The lead ranges from 5 days to about 4.2 years
  - Selling at confirmation would have missed most of the rise

Reading the Real-Time Table

- **Chain**: date-stamped runs joined when the next starts soon after the previous ends
  - At most 93 calendar days later for monthly data (two months without an exceedance), 28 days for weekly data, 10 days for daily GameStop
  - A hand-worked case: A Hand-Worked Date-Stamping Example
- Nasdaq 100: the chain joins 3 runs, from Jun 1995; the strict rule keeps only the run containing the peak
  - Strict rule: start Sep 1998, confirmed Mar 1999, leads 18 and 12 months instead of 57 and 51
  - For BET, Bitcoin and GameStop the two rules give the same start
- Two falls for the same dot-com crash
  - This table: -76%, the lowest daily close within 2 years of the peak
  - Nine Episodes table: -83% within 4 years; the trough of Oct 2002 came 30 months after the peak, outside the 2-year window
- Bitcoin 2021: the peak here is the local high of 13 Apr 2021, not the all-time high of 8 Nov 2021
  - The explosive chain ended in May 2021; the later high came after a deep fall and a slower recovery that BSADF did not flag

Crashes as Critical Points



Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0 (Germany); Wikimedia Commons

- Johansen, Ledoit and Sornette (JLS) Johansen, Ledoit & Sornette (2000)
  - Imitation among traders makes the **hazard rate** of a crash rise as a critical time t_c approaches
  - Hazard rate: the crash probability per unit time, given no crash yet
- Rational investors stay in only if the price grows fast enough
  - The extra growth pays for the rising crash risk
- Result: **faster-than-exponential** growth
  - Decorated by oscillations that speed up near t_c
- t_c is the most probable end of the bubble, not a certain crash date Sornette (2017)

The LPPLS Equation

- $\ln p(t) = A + B\,(t_c - t)^m + C\,(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
  - In words: the log price is a power-law trend towards the critical time plus oscillations whose frequency rises as t_c approaches; $p(t)$: price at time t; t_c: critical time
- A: log price at t_c; $B < 0$: the super-exponential trend for a rising bubble; $0 < m < 1$: power-law exponent
  - The growth rate of the power-law trend, $-Bm(t_c - t)^{m-1}$, rises without bound as $t \to t_c$
- C, ω, ϕ: amplitude, angular log-frequency and phase of the log-periodic oscillations
- Seven parameters: A, B, C, ϕ and the nonlinear t_c, m, ω
- LPPL: the log-periodic power law pattern; LPPLS: the model with the finite-time singularity at t_c

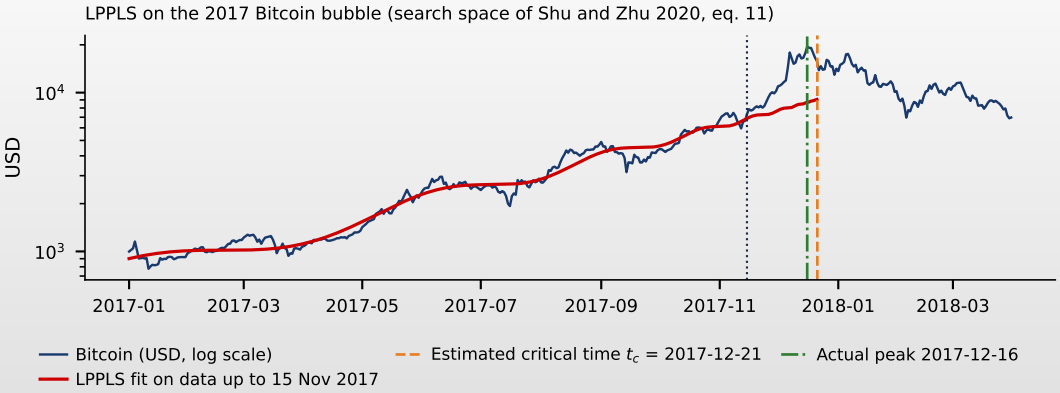
Calibration: Filimonov and Sornette

- Rewrite the cosine: $C\cos(\omega\ln(t_c - t) - \phi) = C_1\cos(\omega\ln(t_c - t)) + C_2\sin(\omega\ln(t_c - t))$ Filimonov & Sornette (2013)
  - $C_1 = C\cos\phi$, $C_2 = C\sin\phi$; A, B, C_1, C_2 are now **linear**: for given (t_c, m, ω) they come from OLS on the window $[t_1, t_2]$
- Search only over (t_c, m, ω): a grid, then a Nelder–Mead refinement (a derivative-free simplex search) of the sum of squared residuals
- Search space of Shu and Zhu (Section 2.3, eq. 11) Shu & Zhu (2020)
  - $m \in [0, 1]$, $\omega \in [1, 50]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/3]$
  - Damping $\frac{m|B|}{\omega|C|} \geq 1$: an approximate condition for a non-negative crash hazard rate
  - Exact condition: $m|B| \geq |C|\sqrt{m^2 + \omega^2}$ (Seminar 17, A8)
- Many local minima remain: different windows and starting points give different t_c Geraskin & Fantazzini (2013)

Qualified Fits: The Filter

- A fit on $[t_1, t_2]$ is **qualified** if it passes all conditions of Shu and Zhu (Section 2.3, eq. 12), and $B < 0$ for a rising bubble Shu & Zhu (2020)
  - $m \in [0.01, 0.99]$, $\omega \in [2, 25]$, $t_c \in [t_2, t_2 + (t_2 - t_1)/5]$
  - Damping $D = m|B|/(\omega|C|) \geq 1$: the search constraint is also required by the qualification filter here
  - Number of oscillations (half-periods) $O = \frac{\omega}{\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$, so the oscillations are not noise
  - Largest relative error of the fitted price $\max_t |\hat{p}_t - p_t|/p_t \leq 0.15$
  - Lomb periodogram of the detrended residual $(t_c - t)^{-m}(\ln p_t - A - B(t_c - t)^m)$: log-periodicity significant at 10%
  - Lomb periodogram: spectral power at each frequency for unevenly spaced points, here the values of $\ln(t_c - t)$; significant: a spectral peak that noise alone would rarely produce
  - Residual unit-root check for $\ln \hat{p}_t - \ln p_t$: Dickey–Fuller and Phillips–Perron reject a unit root at 10% (a stationary, mean-reverting residual; this alone does not make it an autoregressive AR(1) process)
  - Phillips–Perron: a Dickey–Fuller-type unit-root test with a nonparametric correction for serial correlation
- $[t_1, t_2]$: the estimation window; the filter is applied to each window separately

LPPLS on Bitcoin, 2017 (1/2)



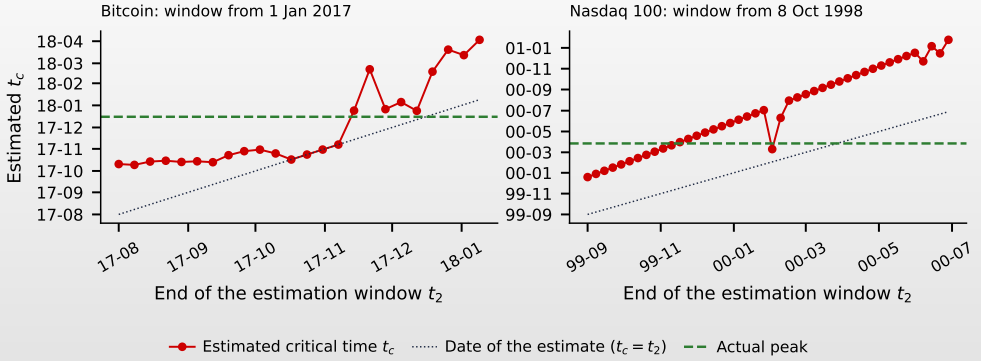
MFM_ch17_lppls

LPPLS on Bitcoin, 2017 (2/2)

- Fit on daily log prices, 1 January to 15 November 2017
  - The path after 15 November is out of sample
- The fit follows the 2017 rise and its oscillations closely
  - $m = 0.95$, $\omega = 10.54$, $B = -2.62$
- Estimated t_c: 21 Dec 2017; actual peak: 16 Dec 2017
  - The peak came 5 days earlier
- The fit fails the condition on relative error ≤ 0.15
  - It does **not** pass the filter, even though it looks convincing

MFM_ch17_lppls

How Stable Is the Critical Time? (1/2)



MFM_ch17_lppls

How Stable Is the Critical Time? (2/2)

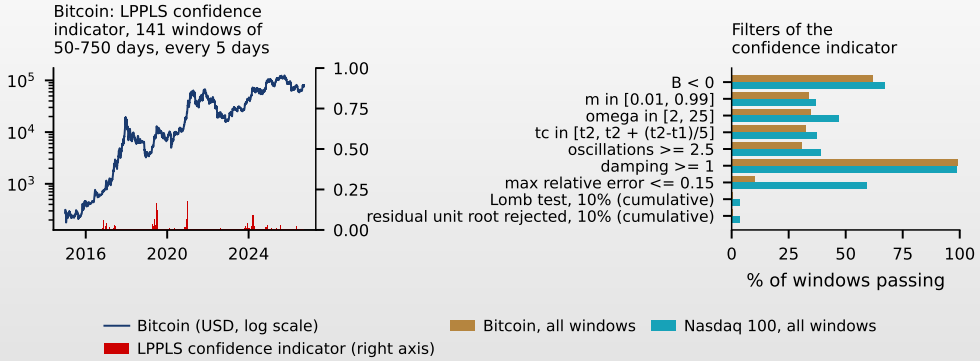
- A good in-sample fit is not a forecast
  - Each point: one fit on a window with a fixed start, ending on the date on the horizontal axis (every 7 days)
- Bitcoin: before the peak (20 fits) t_c ranged from Oct 2017 to Feb 2018
  - Mean absolute error 48 days; within 30 days of the peak: 20% of the fits
- Nasdaq 100: before the peak (30 fits) t_c ranged from Dec 1999 to Sep 2000
  - Mean absolute error 70 days; $t_c \geq t_2$ is a constraint of the search, not a forecast
- Qualified fits before the peak: 0% (Bitcoin), 0% (Nasdaq 100)
  - The birth of a bubble is easier to diagnose than its burst Demos & Sornette (2017)

MFM_ch17_lppls

The LPPLS Confidence Indicator

- At each date t_2, fit many windows $[t_2 - L, t_2]$ and report the **share of qualified fits** Demirer et al. (2019)
  - Grid of Shu and Zhu (Section 2.3): 141 windows, L from 750 down to 50 observations in steps of 5; t_2 every 5 observations; data up to t_2 only Shu & Zhu (2020)
  - Daily closes; Bitcoin trades every day, so one observation is one calendar day
- A high share means the bubble signature is robust to the choice of window
- Honest evaluation: is a signal followed by a fall of at least 20% within 90 days more often than no signal?
  - Fall measured from the price on the signal date; only dates whose 90 calendar days are fully observed (the last three months of data are dropped)

Bitcoin: Real-Time Confidence Indicator (1/2)



MFM_ch17_lppls_confidence

Bitcoin: Real-Time Confidence Indicator (2/2)

- Left: Bitcoin price and, as bars on the right axis, the share of the 141 windows with a qualified fit, every 5 days, 2016–2026
  - Right: share of all windows passing each condition, Bitcoin and, for comparison, the Nasdaq 100
  - Cumulative rows: the Lomb and residual tests count only windows that pass every condition above them
- Positive on 14% of the 2017 dates, 3% in 2021, 12% in other years
  - Maximum 0.18 in Dec 2020
- A 20% fall within 90 days followed 22% of signals, 37% of no-signal dates
  - Bitcoin: 709 dates; Nasdaq 100, 1997–2026: 4% vs 11% (1482 dates)
- The raw difference is not yet evidence (test on the next slide)
  - Signals come in runs and the 90-day outcome windows overlap

MFM_ch17_lppls_confidence

Is the Indicator Better than Chance? A Test (1/2)

- H_0: $P(\text{fall} \geq 20\% \text{ within 90 days} \mid \text{signal}) = P(\cdot \mid \text{no signal})$
  - Statistic $\Delta = \hat{p}_{\text{on}} - \hat{p}_{\text{off}}$: share of signal dates followed by such a fall minus the share of no-signal dates
- Overlap: dates 5 days apart share up to 85 of their 90 outcome days
  - The two-sample z-test assumes independent dates and so understates the standard error
- Dependence-robust inference
  - Circular block bootstrap of the (signal, outcome) pairs: blocks of 90 days, 9 999 replications
  - Circular block bootstrap: resample blocks of consecutive dates, wrapping from the end of the sample to its start, so the dependence inside a block is kept
  - Linear probability regression of the outcome on the signal, with Newey–West standard errors
  - Newey–West: a standard error robust to autocorrelation up to a chosen lag, here 18 dates (90 days at 5-day spacing)

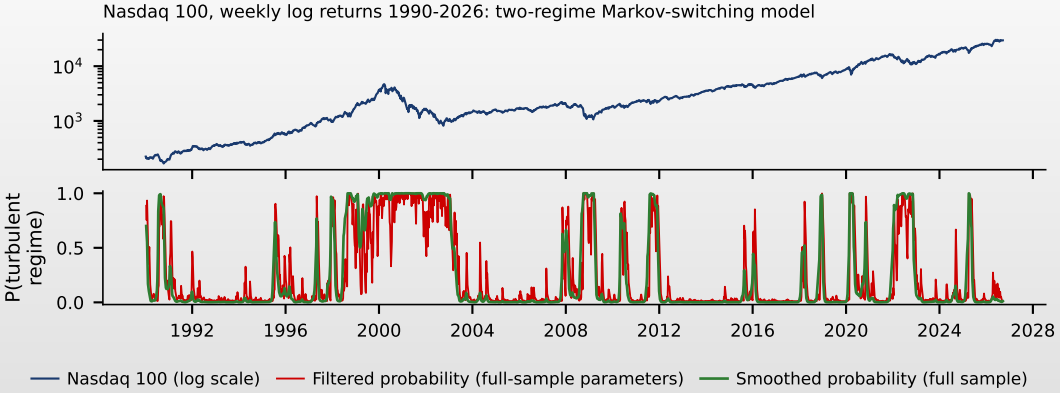
Is the Indicator Better than Chance? A Test (2/2)

- Bitcoin: $\Delta = -15.5$ points, 95% CI $[-30.0, 1.8]$; Newey–West two-sided $p = 0.049$ for $H_0 : \Delta = 0$
  - Block SE 8.1 vs independent-data SE 5.0
  - The percentile CI and the Newey–West t-test use different standard errors, so they need not agree near the 5% boundary
  - The two-sided p counts deviations in both directions; one-sided for a **lower** rate after signals, $p = 0.024$, a direction not stated in advance
  - The same test for the Nasdaq 100, with a power calculation: Seminar 17, Exercise B7
- Weak evidence: the block-bootstrap CI contains zero; the Newey–West regression is borderline (two-sided $p = 0.049$)
  - The point estimate is negative: if anything, a lower crash rate after signals
  - Signals come in steady rises, not just before crashes

The Markov-Switching Model

- Weekly log return $r_t = \mu_{s_t} + \sigma_{s_t}\varepsilon_t$, $\varepsilon_t \sim N(0,1)$, regime $s_t \in \{1,2\}$ unobserved Hamilton (1989)
  - $r_t = 100\ln(P_t/P_{t-1})$ in %, a log return, not the simple change $100(P_t/P_{t-1}-1)$
  - Each regime has its own mean μ and volatility σ: a calm and a turbulent regime
- Regimes follow a Markov chain: $p_{ij} = P(s_t = j \mid s_{t-1} = i)$; expected duration of regime i: $1/(1-p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter, which updates $P(s_t = j \mid r_1, \ldots, r_t)$ week by week
- **Filtered** probability: returns up to t for given parameters; **smoothed**: the whole sample (hindsight)
  - Our parameters are estimated on the full sample, so our filtered probabilities are only **pseudo real time**; genuine real time needs parameters re-estimated with data up to t
  - A worked filter step is in Seminar 17, Exercise A2

Nasdaq 100: Two Regimes (1/2)



MFM_ch17_markov_switching

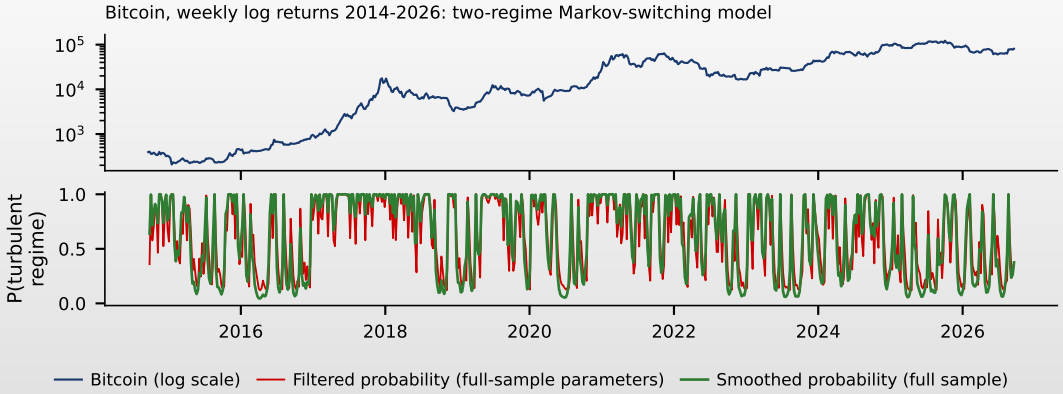
Nasdaq 100: Two Regimes (2/2)

Regime	$\hat{\mu}$ (SE)	$\hat{\sigma}$	$\hat{p}_{ii}$	Dur.
Calm	0.43 (0.06)	2.30	0.987	75.4
Turbulent	−0.32 (0.27)	5.40	0.957	23.5

- Weekly log returns in %, 1990–2026, 1 915 weeks
  - Duration in weeks; calm regime in 76% of the weeks
- Turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5)
  - Filtered probability 0.99 at the end of March 2000: prices still rising, volatility jumping

MFM_ch17_markov_switching

Bitcoin: Two Regimes (1/2)



MFM_ch17_markov_switching

Bitcoin: Two Regimes (2/2)

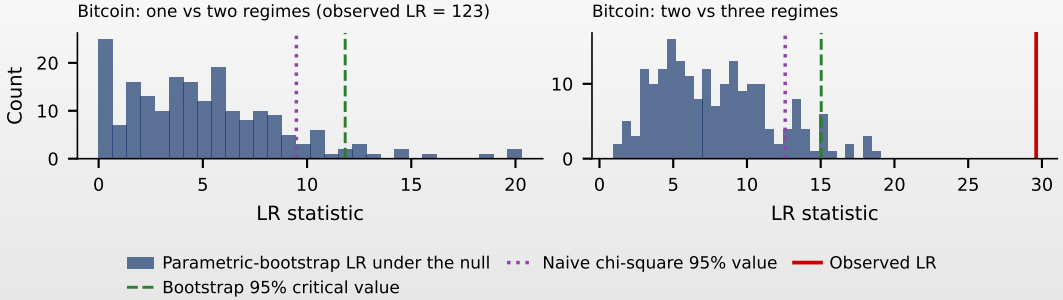
- Weekly log returns 2014–2026, 626 weeks
- Calm: $\hat{\mu} = 0.26\%$ per week, $\hat{\sigma} = 2.93\%$, duration 3.8 weeks
  - Turbulent: $\hat{\mu} = 1.20\%$ (SE 0.58), $\hat{\sigma} = 11.22\%$
  - Turbulent duration 6.3 weeks, 62% of the weeks
- Unlike equities, the turbulent regime has the **higher** mean
  - Regimes are short: volatility states, not bubble phases
- Regimes ignore the price level: a slow, calm bubble stays calm
  - BSADF says "the price is accelerating"; the regime says "risk has changed"

MFM_ch17_markov_switching

How Many Regimes? A Bootstrap LR Test (1/3)

- H_0: one regime (iid, independent and identically distributed, Normal returns) vs H_1: two Markov regimes
  - Statistic $\mathrm{LR} = 2(\hat{\ell}_2 - \hat{\ell}_1)$; $\hat{\ell}_1$, $\hat{\ell}_2$: maximised log-likelihoods with one and two regimes
  - A large LR favours two regimes
- Non-standard null distribution (Appendix) Hansen (1992)
  - p_{11}, p_{22} are not identified under H_0, so LR is not χ^2
- **Parametric bootstrap**
  - Simulate from the fitted H_0 model, refit both models, recompute LR; $B = 199$ replications
  - $p = (1 + \#\{\mathrm{LR}^* \geq \mathrm{LR}\})/(B + 1)$; $\#\{\cdot\}$: number of bootstrap values LR^* at least as large as the observed LR

How Many Regimes? A Bootstrap LR Test (2/3)



MFM_ch17_inference

How Many Regimes? A Bootstrap LR Test (3/3)

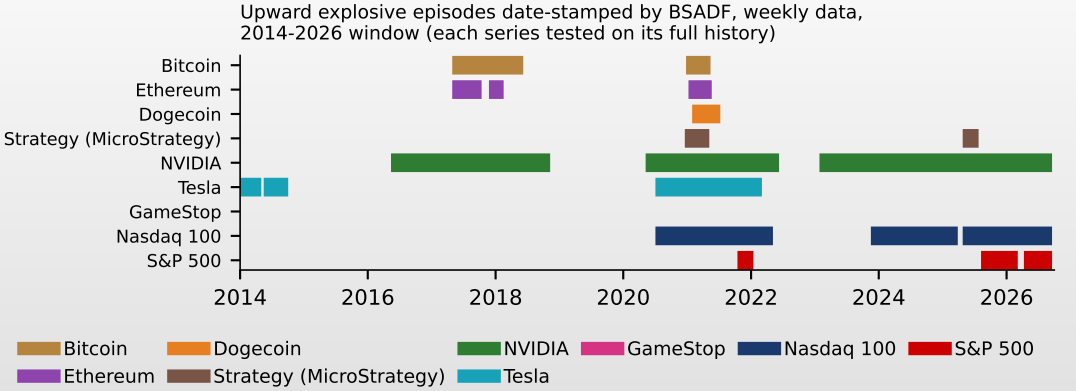
- One vs two regimes
  - Bitcoin: LR $= 123.2$ (95% value 11.8, $p = 0.005$); Nasdaq 100: LR $= 475.7$ (11.1, $p = 0.005$)
- Two vs three regimes, Bitcoin (replication target of Seminar 17, B6)
  - LR $= 29.6$; bootstrap 95% value 15.0 vs naive χ^2_6 value 12.6; $p = 0.005$
- One regime is rejected for both series; a third Bitcoin regime is also supported
  - Caveat: the one-regime null is Normal, so fat tails alone can favour two regimes

MFM_ch17_inference

Are Crypto-Assets Bubbles?

- No dividends: the fundamental value of Bitcoin is hard to define; one estimate puts it at zero Cheah & Fry (2015)
- Alternative: value from network adoption (Metcalfe's law), with LPPL bubbles on top Pele & Mazurencu-Marinescu-Pele (2019)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble Pele et al. (2023)
- Without a fundamental value we can still test the price: is it explosive, and when?

Explosive Episodes in Crypto and Tech Stocks (1/2)



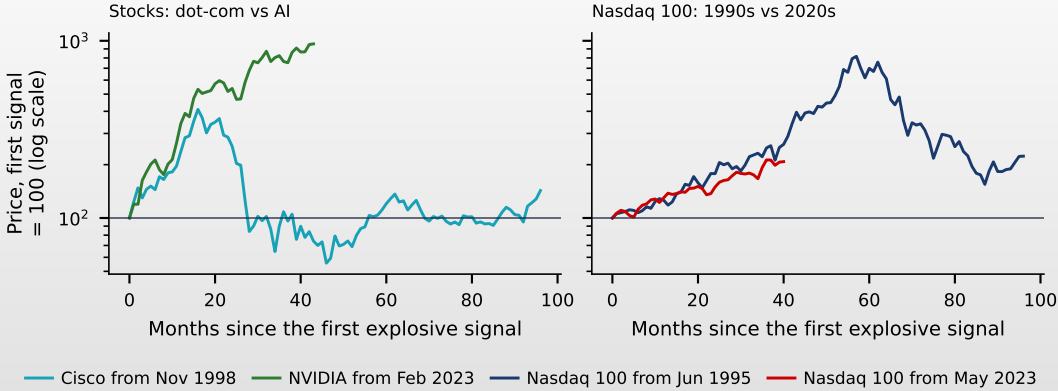
MFM_ch17_crypto_ai

Explosive Episodes in Crypto and Tech Stocks (2/2)

- Weekly log closes; rising episodes only
  - Each series tested on its own history, with its own Monte Carlo critical values
- Crypto booms cluster in 2017 and late 2020–2021
  - Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021
- Stocks
  - Strategy (formerly MicroStrategy; holds Bitcoin): Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022; NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
  - GameStop, weekly: the 3-week surge of January 2021 is shorter than the minimum duration
  - Indices: Nasdaq 100 Jul 2020–May 2022; Nov 2023–Mar 2025; Apr 2025–ongoing; S&P 500 Oct 2021–Jan 2022; Aug 2025–Mar 2026; Apr 2026–ongoing
- An accelerating **fall** also gives $b > 0$: always report the direction

MFM_ch17_crypto_ai

Dot-Com vs AI (1/2)



MFM_ch17_crypto_ai

Dot-Com vs AI (2/2)

- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)
  - First signal: start of the first date-stamped run beginning in 1995 or later (dot-com) or in 2023 or later (AI), not of the run that contains the peak
- Cisco: first signal Nov 1998; NVIDIA: first signal Feb 2023
  - Cisco peaked after 16 months at 4.1 times the signal price
  - NVIDIA, 43 months later: 9.6 times, episode still open
- Nasdaq 100: the AI rise is slower than in the 1990s
  - From Jun 1995: peak after 57 months at 8.2 times; from May 2023: 2.1 times after 40 months

MFM_ch17_crypto_ai

Is There an AI Bubble? What the Tests Can Say

- Monthly BSADF flags NVIDIA and the Nasdaq 100
  - NVIDIA: Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing; Nasdaq 100: current episode from May 2023
  - Monthly dates ($T = 332$, $w_0 = 36$, $L = 6$ months); the weekly timeline ($T = 1\,443$, $w_0 = 82$, $L = 8$ weeks) shifts the dates by up to a few months, because T, w_0 and L change with the frequency
- The tests cannot separate a bubble from explosive fundamentals Pástor & Veronesi (2009)
  - Fast-growing expected profits of a new technology can also make prices explosive
- History
  - Cisco's dividend-adjusted share price returned to its March 2000 level only in Aug 2021
- What we can say
  - The tests detect explosive price dynamics, not their cause
  - The industry run-up evidence Greenwood, Shleifer & You (2019) motivates watching crash risk, but does not quantify it for these stocks
  - The timing of any crash is unknown

Case Study (1/8): the Research Question

- Paper: DeFusco, Nathanson & Zwick (2022), *Journal of Financial Economics*, 146(1), 205–229
  - Setting: the US housing cycle of 2000–2011, the boom and bust behind the 2008 crisis
- Questions (Introduction)
  - Who buys during a boom?
  - Can speculation explain the joint path of prices, transaction volume and unsold listings?
- Speculation, in the Harrison–Kreps definition the authors adopt (footnote 1)
  - Paying more for an asset because one has the right to resell it
- Three phases (Section 2 of the paper)
  - Boom: prices and volume rise
  - "Quiet": volume falls while prices still rise and unsold listings pile up
  - Bust

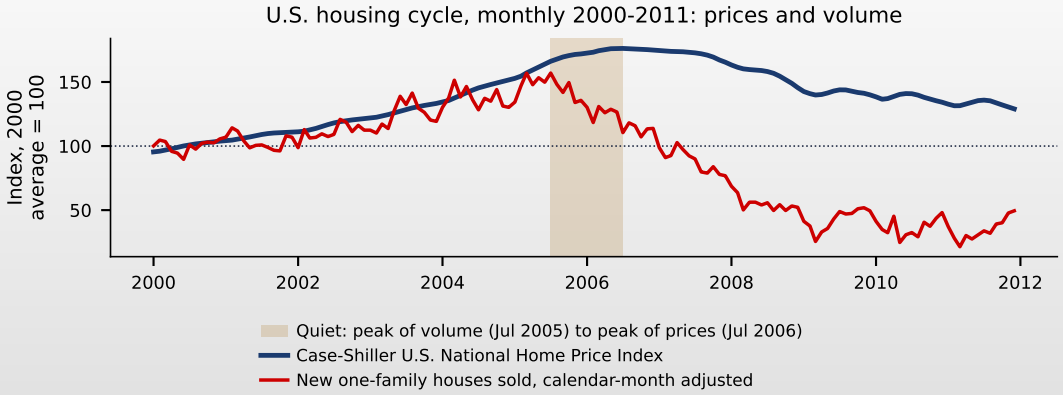
Case Study (2/8): Why It Matters for This Chapter

- The explosiveness tests of this chapter use prices only
  - Bubbles for Fama also study turnover, issuance and volatility Greenwood, Shleifer & You (2019)
- Bubbles come with high turnover Scheinkman & Xiong (2003)
  - They trade the resale option of Bubbles Beyond Rational Expectations
- Quantities show who drives a boom
  - Blockchain flows did so for the 2017 Bitcoin boom Griffin & Shams (2020)
- This paper adds the housing evidence
  - Who the speculators are, when they trade and how much of the cycle they explain

Case Study (3/8): Data, Method and Headline Results

- Data (Section 1.1): 51 580 408 home sales from deeds records, 1995–2014, 115 metropolitan areas
  - **Speculators** (Section 1.2): short-term buyers who resell within three years, and non-occupant buyers
- Lead–lag (Eq. 1): $p_{i,t} = \beta_k v_{i,t-k} + \eta_{i,t}$, $k = -12, \ldots, 48$ months
  - i: metropolitan area; t: month; p: demeaned log price; v: demeaned sales rate; $\eta_{i,t}$: error
  - $k > 0$: volume k months earlier; implied correlation $\beta_k \, \mathrm{sd}(v_{i,t-k})/\mathrm{sd}(p_{i,t})$, sd: standard deviation
- Figures 2 and 4
  - The correlation peaks at a lag of 24 months
  - Short-term sales explain 39% of the 2000–2005 rise in sales
- Table 2: cities with more speculation had larger booms and larger busts
  - One standard deviation more speculative volume: boom larger by 24.9 points, bust deeper by 7.4 points
- Model (Sections 5–6, Table 6): extrapolating short-horizon buyers amplify the cycle
  - A 5% purchase tax cuts the bust from 8.2% to 1.1%

Case Study (4/8): Prices and Volume, 2000–2011



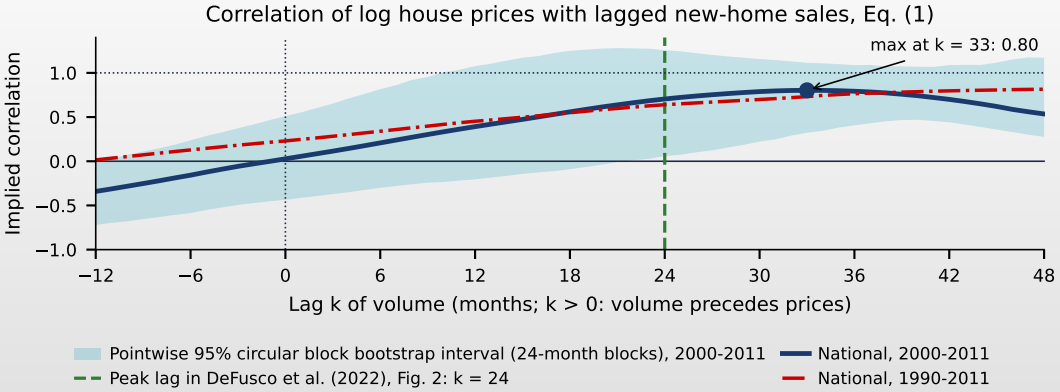
MFM_ch17_case_study

Case Study (5/8): Prices and Volume, 2000–2011

- FRED (Federal Reserve Economic Data, St. Louis Fed): Case-Shiller U.S. National Home Price Index and new one-family houses sold
  - 2000 average = 100; volume adjusted by calendar-month means, as in Figure 1 of DeFusco, Nathanson & Zwick (2022)
- Boom: sales peak at 157 in Jul 2005
  - Paper: 150% of the 2000 level; prices keep rising to 176 in Jul 2006
- Quiet: 12 months from the volume peak to the price peak
  - Sales already down to 110 at the price peak (unadjusted sales: 16 months)
- Bust: 2011 averages 35 for sales, 133 for prices
  - New-home sales exclude resales, so they cannot show who trades

MFM_ch17_case_study

Case Study (6/8): Volume Leads Prices



MFM_ch17_case_study

Case Study (7/8): Volume Leads Prices

- Eq. (1) on the national series, without a constant
  - Pointwise 95% band: circular block bootstrap, 24-month blocks, 2000 replications
- 2000–2011: maximum at $k = 33$ (0.80, CI $[0.32, 1.11]$)
  - At the paper's $k = 24$: 0.70, CI $[0.06, 1.26]$
- The band excludes zero for $k = 22, \ldots, 48$
  - But one national cycle holds few independent blocks: wide intervals
- Direction as in the paper: volume first, by two to three years
  - The length is not precise (1990–2011: $k = 48$)

MFM_ch17_case_study

Case Study (8/8): Limits and Open Questions

- Data limit of our illustration (Seminar 17, C3)
  - The public series are national and cover new homes only: short-term and non-occupant volume cannot be built
  - Figures 1–2 can be illustrated, Table 2 cannot
- Limits stated by the authors
  - Holding periods are realised, not intended (footnote 4; instrumental-variable check in the Online Appendix)
  - The non-occupant proxy misclassifies owners who give the property address for tax mail (Section 1.2)
  - Selling horizons are exogenous in the model, so the counterfactuals may overstate short-term buyers (Section 4.3)
  - Credit and new construction are left out (Sections 6.5.3, 7)
- Open questions
  - Does trading volume also lead prices in crypto booms?
  - Is there a quiet (falling volume, rising prices) before the end of the Bitcoin episodes dated by BSADF?

Can Bubble Alarms Warn of Crashes?

Real-time bubble alarms

- Questions
  - Can bubble alarms be raised in real time with a controlled false-alarm rate?
  - Do they carry crash information?
- BSADF date-stamping: one right-tailed test per date
  - Size controlled date by date only (Size of GSADF When Volatility Changes: Results)
  - Live research: monitoring rules for bubbles *and* crashes Whitehouse, Harvey & Leybourne (2023); "econometric thermometers" for housing Shi & Phillips (2023)
- Formal version: alarm $A_t \in \{0, 1\}$, crash $C_{t,h}$ = a fall of at least 20% within $h = 90$ days
  - H_0: $P(C_{t,h} \mid A_t = 1) = P(C_{t,h} \mid A_t = 0)$; H_1: larger after an alarm
  - Falsified if the hit rate after alarms does not exceed the rate without alarms, with inference robust to overlapping windows
- Identification: explosive prices need not be a bubble
  - Explosive expected cash flows of a new technology leave the same signature Pástor & Veronesi (2009)
  - Run-ups raise crash risk without predictable average losses Greenwood, Shleifer & You (2019)

The Discovery Loop and Its Checks

- **Literature**
  - Ask for papers on the false-alarm rate of sequential bubble monitoring
  - Resolve every DOI (Digital Object Identifier, the permanent identifier of a paper) on Crossref; find the table the answer quotes
- **Hypotheses**, each with its own observable implication
  - Rational bubble, explosive expected cash flows, falling discount rate
  - E.g. an explosive price-to-earnings ratio vs explosive earnings
- **Code**
  - Replicate a known number first (GSADF $= 3.85$ for weekly Bitcoin), then extend
- **Robustness** and **critique**
  - Pre-register the grid of horizons and crash thresholds, with a Holm correction
  - Holm: the k-th smallest of m p-values is compared with $0.05/(m-k+1)$, stopping at the first non-rejection
  - Ask the assistant to attack the result as a referee
- **Checks** the researcher must run (details and prompts in the Appendix)
  - No full-sample information in a date stamp; right-tail critical values; wild bootstrap under changing volatility
  - Multiplicity over dates and assets; inference robust to overlapping outcome windows

Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal
  - "A positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher's exact test"
  - Alarms come in runs and outcomes overlap, so we test by circular rotations of the alarm series
- Hit rates after alarms and without: Bitcoin: Real-Time Confidence Indicator
  - Bitcoin: difference -15.5 percentage points; one-sided $p = 0.96$ for $H_1 : \Delta > 0$ (larger after an alarm) from 658 rotations
  - Nasdaq 100: difference -7.1 points, $p = 0.98$
- Pre-registered grid of 4 horizons $\times$ 3 thresholds
  - Smallest raw p: 0.76 (Bitcoin) and 0.69 (Nasdaq 100)
  - No cell rejects H_0 at 5%, before or after the Holm correction
- Interpretation (assumptions in the Appendix)
  - The indicator dates accelerating rises, but these tests do not support a higher probability of large falls after a signal

Your Turn: a Project Seed

- **Questions**
  - Can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period?
  - Can it still warn before large falls?
- **Replicate first**
  - GSADF $= 3.85$ for weekly Bitcoin
  - The crash probability after run-ups above 100% in Greenwood, Shleifer & You (2019) (ours: 46%)
- **Extend**
  - Calibrate the alarm threshold by simulation over the whole monitoring period Whitehouse, Harvey & Leybourne (2023)
  - Evaluate the alarms with rotations or a block bootstrap
- **Deliverables**
  - A pre-registered grid, fixed in a dated document before the evaluation
  - A notebook that regenerates every number; false-alarm and hit rates with valid p-values
- **AI log** (templates)
  - `AI_USE.md`: the prompts that shaped the design
  - `AI_ERRORS.md`: each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it

Key Takeaways

- A bubble is price minus fundamental value
  - Rational bubbles must be explosive; bursting bubbles hide from full-sample tests
- GSADF tests for explosive episodes; BSADF date-stamps them in real time
  - Late volatility rise: Monte Carlo critical values reject 32.0% of true nulls, the wild bootstrap 4.1%
  - Date-stamping over many dates needs family-wise control
- Across our episodes the tests warned months before the peak, but did not time the crash
- LPPLS describes the acceleration
  - Its critical time moves with the window
  - The confidence indicator is judged by a dependence-robust test of hit rates
- Sharp run-ups raise crash risk without a statistically established average loss
  - A risk-management signal, not a trading rule

Which Tool When?

Question	Tool
Has there been any explosive episode?	GSADF with simulated critical values
When did it start and end?	BSADF date-stamping, minimum duration $\lceil \ln T \rceil$; family-wise critical values for long monitoring
Volatility changes a lot (crypto, crises)	Wild bootstrap critical values
Is the rise accelerating super-exponentially?	LPPLS confidence indicator over many windows
Has the risk state changed?	Markov-switching filtered probabilities
How likely is a crash after a run-up?	Run-up conditioning across many assets (Greenwood, Shleifer and You 2019)

Check Yourself, and Next: Chapter 18

Retrieval questions

- Why does a full-sample right-tailed ADF miss a bubble that collapses periodically? revisit
- Why is the 95% critical value of GSADF far above that of a single ADF? revisit
- Why is a single LPPLS critical time not a crash forecast? revisit
- Why must hit rates after alarms be tested with a block bootstrap or rotations? revisit

Chapter 18: systemic risk

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- How much one institution adds to the risk of the system
  - CoVaR: the value-at-risk (VaR) of the system when an institution is in distress
  - SRISK: the capital shortfall of an institution in a crisis
- Further reading: Franke, Härdle & Hafner (2019), Kindleberger & Aliber (2005), Shiller (2015), Sornette (2017)

Transversality: Where the Bubble Term Comes From

- Substitute the pricing equation into itself h times: $P_t = \sum_{i=\mathbf{1}}^{h} \frac{\mathbb{E}_t[D_{t+i}]}{(1+r)^i} + \frac{\mathbb{E}_t[P_{t+h}]}{(1+r)^h}$ (law of iterated expectations)
- Let $h \to \infty$: $P_t = F_t + B_t$ with $B_t = \lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h$
- **Transversality condition**: $\lim_{h\to\infty} \mathbb{E}_t[P_{t+h}]/(1+r)^h = 0$; it holds $\Leftrightarrow B_t = 0 \Leftrightarrow P_t = F_t$
- Shift the limit one period: $\mathbb{E}_t[B_{t+\mathbf{1}}] = \lim_h \mathbb{E}_t[P_{t+\mathbf{1}+h}]/(1+r)^h = (1+r)\,B_t$, the condition that the discounted bubble is a martingale
- Seminar 17, Exercise A1 derives the general solution and why $B_t \geq 0$ rules out a bubble that starts from zero Diba & Grossman (1988)

Back

Bubbles for Fama: Effective Sample Size

- The run-ups fall in 31 calendar years and overlap: consecutive 2-year windows of the same boom share months
- Year-block bootstrap SE (9.1 percentage points) vs binomial SE (6.9): a design effect (squared ratio of the two SEs) of 1.7, i.e. about 30 independent observations
- Blocks of 2, 3 and 5 consecutive years (2-year windows of adjacent years overlap): SE $= 8.9, 8.7, 8.3$
- Unconditional mean 2-year return over all industry-months: 28%
  - Above the CI of the mean return after a 100% run-up (-14% to 20%): returns after run-ups look lower than usual
  - A descriptive comparison, not a formal test of the difference of two dependent estimates

Back

Why the Full-Sample ADF Fails: the Leverage of Collapses

- OLS slope: $\hat{b} = \sum_t \tilde{y}_{t-\mathbf{1}}\,\Delta y_t \big/ \sum_t \tilde{y}^{\mathbf{2}}_{t-\mathbf{1}}$
  - $\tilde{y}_{t-\mathbf{1}}$: the lagged log price minus its sample mean
  - Expansion: $\Delta y_t > 0$ while $\tilde{y}_{t-\mathbf{1}}$ rises above its mean: positive terms
  - Collapse: the most negative Δy_t at the highest $\tilde{y}_{t-\mathbf{1}}$: a few very large negative terms
- Evans form in logs, $y = \ln B$, collapse to $\delta = e^c$ with probability $1 - \pi$ Evans (1991)
  - For $B \gg \delta$ the survival log growth is $\approx \ln \frac{\mathbf{1}+r}{\pi}$
  - $\mathbb{E}[\Delta y_t \mid y_{t-\mathbf{1}}] \approx \pi \ln \frac{\mathbf{1}+r}{\pi} + (1-\pi)(c - y_{t-\mathbf{1}})$
  - Slope $-(1-\pi) < 0$: apparent mean reversion, although $\mathbb{E}_t B_{t+\mathbf{1}} = (1+r)B_t$
- Windows with the expansion but not the collapse have $\hat{b} > 0$ when the log price accelerates (next slide)
  - A supremum over windows restores the power Phillips, Shi & Yu (2015b)

Back

Explosive in Levels, Accelerating in Logs

- A rational bubble is explosive in the **level**: $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$, an autoregressive root $1+r>1$
  - In logs a pure bubble path $B_t = B_0(1+r)^t$ has constant growth: $\ln B_t = \ln B_0 + t\ln(1+r)$, a deterministic linear trend in logs, $b=0$ in $\Delta y_t = a + b\, y_{t-1} + \varepsilon_t$
- The log price $\ln(F_t + B_t)$ **accelerates** while the bubble share grows: its log growth rises from that of F towards $\ln(1+r)$ (a convex log path)
  - Once the bubble dominates, log growth is roughly constant again: the log-price regression then has little power
- So the log-price test targets a different alternative, **accelerating** log prices, not the level condition itself; the specification of the null and of the deterministic terms matters Phillips, Shi & Yu (2014)
- In this chapter we test log prices and log ratios (like PWY for the Nasdaq); PSY test the price-dividend ratio Phillips, Wu & Yu (2011), Phillips, Shi & Yu (2015a)
- Numerical version on the main path: Levels Explode, Log Prices Accelerate

Back

Limit Theory (1): the Null Distribution

- Null model: $y_t = d\,T^{-\eta} + y_{t-1} + \varepsilon_t$, $\eta > 1/2$, ε_t iid $(0, \sigma^2)$ Phillips, Shi & Yu (2015a)
  - A random walk with an asymptotically negligible drift $d\,T^{-\eta}$ (d: a constant)
- Limit (PSY, Theorem 1, eq. 5), by the functional central limit theorem and continuous mapping
  - $$\mathrm{ADF}_{r_1}^{r_2} \Rightarrow \frac{\frac{1}{2} r_w \left[W(r_2)^2 - W(r_1)^2 - r_w\right] - \int_{r_1}^{r_2} W\,dr\,\left[W(r_2) - W(r_1)\right]}{r_w^{1/2} \left\{ r_w \int_{r_1}^{r_2} W^2 dr - \left(\int_{r_1}^{r_2} W\,dr \right)^2 \right\}^{1/2}}$$
  - $r_w = r_2 - r_1$; W: standard Brownian motion; $\Rightarrow$: convergence in distribution as $T \to \infty$
- σ and d disappear: the limit is pivotal
  - SADF and GSADF are suprema of the same functional: their critical values depend only on r_0 and the deterministic terms
- $r_1 = 0$, $r_2 = 1$: the Dickey–Fuller limit with a constant
  - Its 95% quantile is negative (simulated: -0.06), far from 1.645
  - The drift and intercept specification changes the limit and the size Phillips, Shi & Yu (2014)

Back

Limit Theory (2): Mildly Explosive Alternatives

- Alternative: $y_t = \rho_T\, y_{t-1} + \varepsilon_t$ with $\rho_T = 1 + c/k_T$, $c > 0$, $k_T \to \infty$, $k_T = o(T)$; PSY use $k_T = T^{\alpha}$, $\alpha \in (0, 1)$
  - c: how explosive; k_T: the speed at which the root approaches 1; $k_T = o(T)$: $k_T / T \to 0$
  - Closer to 1 than any fixed explosive root, farther than a local-to-unity root $1 + c/T$: the "mild" speed of real bubbles
- Phillips and Magdalinos: $\dfrac{\rho_T^T}{\rho_T^2 - 1}\,(\hat{\rho}_T - \rho_T) \Rightarrow \mathcal{C}$, a standard Cauchy law, without Gaussian errors Phillips & Magdalinos (2007)
  - $\hat{\rho}_T$: the OLS estimate of ρ_T; $\mathcal{C}$: the standard Cauchy distribution
  - Rate $k_T \rho_T^T$: exponentially fast; the ADF statistic diverges to $+\infty$
- Hence SADF and GSADF diverge too (for GSADF any window of at least $r_0 T$ observations inside the bubble suffices): the tests are consistent, power $\to 1$ Phillips, Shi & Yu (2015b)
- Finite samples: power depends on c, α and the duration; short bursts may stay below the bar, as in the Evans simulation

Back

Limit Theory (3): Consistent Date-Stamping

- Crossing times (PSY, eqs. 7–8)
  - $\hat{r}_e = \inf\{r_2 \geq r_0 : \mathrm{BSADF}_{r_2} > scv_{r_2}^{\beta_T}\}$, $\hat{r}_f = \inf\{r_2 \geq \hat{r}_e + \delta \ln(T)/T : \mathrm{BSADF}_{r_2} < scv_{r_2}^{\beta_T}\}$
  - $\hat{r}_e$, $\hat{r}_f$: estimated start and end, as fractions of the sample; inf: the first r_2 that meets the condition
  - $scv_{r_2}^{\beta_T}$: the 100$(1-\beta_T)$% quantile of BSADF on $\lfloor Tr_2 \rfloor$ observations; β_T: the level; $\lfloor \cdot \rfloor$: rounded down
- Consistency (PSY, eq. 11): $(\hat{r}_e, \hat{r}_f) \to_p (r_e, r_f)$ if $\dfrac{1}{scv^{\beta_T}} + \dfrac{scv^{\beta_T}}{T^{1-\alpha/2}} \to 0$
  - $\to_p$: convergence in probability to the true dates r_e, r_f
  - The critical value must diverge, so that false alarms vanish ($\beta_T \to 0$)
  - But slower than $T^{1-\alpha/2}$, so that true bubbles are still found; e.g. like $\ln T$ Phillips, Shi & Yu (2015a)
- In practice β_T is fixed at 5% at every date
  - Over hundreds of dates a false alarm becomes likely (the Monte Carlo size study in the wild bootstrap section)

Back

The Price-Dividend Ratio: Comparison with PSY

- Our critical values: 1 000 Monte Carlo paths of the sample length
- PSY, on data to 2010 and without our filter of 8 consecutive months, also date 1879–1880, 1955–1956 and 1986–1987 Phillips, Shi & Yu (2015a)
- With a shorter minimum window 1879–1880, 1954–1956 and 2008–2009 appear in our data too (Seminar 17, B1)

Back

Family-Wise Critical Values: Bootstrap Design

- Phillips and Shi: the 95% quantile of the maximum of BSADF over τ_b consecutive dates Phillips & Shi (2020)
  - Sets the FWER at 5% for the bootstrap design: τ_b monitoring dates after a start-up of w_0 observations
- Resampling
  - Residuals e_t of $\Delta y_t = a + e_t$, drawn with replacement and multiplied by $N(0,1)$ weights
  - Paths of $w_0 + \tau_b - 1$ changes; 999 replications
- Why the control is only approximate here
  - BSADF over the full history: later dates search more start points than the bootstrap design
  - Drawing with replacement loses the time order: unlike the pointwise wild bootstrap, which keeps each Δy_t at its date, these paths ignore the volatility clustering of Bitcoin
- The pointwise rule gives a crossing within the last 104 weeks in 42% of null paths

Back

Real Time vs Hindsight: Caveats

- Column definitions as on the main slide, with three caveats
  - Confirmed is only pseudo real time: w_0 and L come from the full-sample T (a truly real-time design fixes them ex ante)
  - GameStop: leads in calendar days converted to months
  - GameStop: fall to the lowest close within 1 year, not 2
- Bitcoin 2021 and the two Nasdaq 100 falls: explained on the main slide Reading the Real-Time Table

Back

Real-Time Monitoring: Newer Procedures

- Repeating a one-shot test every period inflates the false-positive rate Astill et al. (2018)
  - Astill et al. compare monitoring statistics with the maximum over a training period
  - The false-positive rate is then set for a chosen horizon, robust to time-varying volatility
- Bubbles and crashes Whitehouse, Harvey & Leybourne (2023)
  - After a bubble is detected, monitor for the crash with the sign of the mean of Δy_t
  - It never signals a crash before the crash starts
- Earlier detection Whitehouse, Harvey & Leybourne (2025)
  - Alternative variance standardisations keep the false-detection probability but find bubbles sooner (Bitcoin, housing)
- Collapse phase: PSY applied to the reversed sample dates the crisis and the recovery Phillips & Shi (2018)
- Alternatives to sup-ADF
  - A Chow-type break test, random walk to explosive, has the highest power in Homm & Breitung (2012)
  - Recursive tests with a start-point search Harvey, Leybourne & Sollis (2015)

Back

LPPL in the Cryptocurrency and Romanian Markets

- Bucharest Stock Exchange: herding and log-periodic patterns before the 2007–2008 crash Pele, Mazurencu & Nijkamp (2013); crash modelling for the BVB Pele & Mazurencu-Marinescu (2012)
- Cryptocurrencies: a fundamental value from Metcalfe's law (network size) combined with LPPL bubbles Pele & Mazurencu-Marinescu-Pele (2019), Conda et al. (2026)
- Dogecoin 2021: LPPL together with sentiment from social media Pele (2025)
- Common lesson: LPPL describes the acceleration well; the crash date remains the weak point

Back

Estimation: Hamilton Filter, Kim Smoother, EM

- Log-likelihood of θ (all μ_j, σ_j, p_{ij}), by prediction and update Hamilton (1989)
  - $\ell(\theta) = \sum_t \ln \sum_j P(s_t = j \mid \mathcal{F}_{t-1})\, \varphi(r_t; \mu_j, \sigma_j^2)$
  - $\mathcal{F}_t$: returns up to t; $\varphi(x; \mu, \sigma^2)$: the Normal density with mean μ and variance σ^2
- Kim smoother, backwards Kim (1994)
  - $P(s_t = j \mid \mathcal{F}_T) = P(s_t = j \mid \mathcal{F}_t) \sum_k p_{jk} \dfrac{P(s_{t+1} = k \mid \mathcal{F}_T)}{P(s_{t+1} = k \mid \mathcal{F}_t)}$
- EM algorithm: the M-step is a weighted average Hamilton (1990)
  - Smoothed $\xi_{tj} = P(s_t = j \mid \mathcal{F}_T)$, $\xi_{t,ij} = P(s_{t-1} = i, s_t = j \mid \mathcal{F}_T)$
  - $\hat{\mu}_j = \dfrac{\sum_t \xi_{tj} r_t}{\sum_t \xi_{tj}}$, $\hat{\sigma}_j^2 = \dfrac{\sum_t \xi_{tj}(r_t - \hat{\mu}_j)^2}{\sum_t \xi_{tj}}$, $\hat{p}_{ij} = \dfrac{\sum_{t \geq 2} \xi_{t,ij}}{\sum_{t \geq 2} \xi_{t-1,i}}$
- Unbounded likelihood ($\sigma_j \to 0$ on one return) and several local maxima
  - Best interior maximum from 20 random starts; no $\hat{\sigma}_j$ may collapse towards 0, also in bootstrap refits
  - Standard errors from the inverse numerical Hessian assume a known number of regimes

Back

Why the LR Test for Regimes Is Non-Standard

- Under H_0 the regime means and variances coincide Hansen (1992)
  - p_{11}, p_{22} are not identified: nuisance parameters present only under H_1
  - The information matrix is singular and some scores are identically zero
- Hansen: bounds on the standardised LR via empirical-process theory Hansen (1992)
- Carrasco, Hu and Ploberger: an optimal test Carrasco, Hu & Ploberger (2014)
  - It needs only the estimate under H_0, with bootstrap critical values
- Two vs three regimes: the same logic
  - The null model is then the fitted two-regime chain

Back

Case Study, Data and Method

- Data (Section 1.1): arm's-length sales only
  - 115 metropolitan areas (48% of the US housing stock); listings linked to sales in 57 of them
  - The cycle analysis covers 2000–2011: realised holding periods need consecutive sales
- Speculators (Section 1.2)
  - Short-term buyers: resell within three years (realised holding period)
  - Non-occupant buyers: a mailing address different from the property address (102 areas)
- Lead–lag (Eq. 1, Figure 2): construction of the variables
  - p: log price, demeaned by city; v: sales over the 2000 housing stock, demeaned by city and calendar month
  - Standard errors clustered by month
- Cross-section (Table 2): OLS across cities on the speculative volume boom (2000–2005)
  - Outcomes: price boom (2000–2006), price bust (2006–2011), listings, volume, foreclosures
- Model (Sections 5–6): extrapolating buyers with different horizons; not every listing sells
  - Counterfactuals: rational expectations, no speculators, transaction taxes

Back

Case Study, Complete Results (1/2)

- Figure 4: sales rise from 2.73 to 3.82 million (2000 to 2005)
  - Short-term sales: from 510 to 940 thousand, 39% of the rise; non-occupant purchases: 53%
  - Interpretation: speculators account for much of the volume boom
- Table 2, Panel A: price-boom coefficient 1.930 (standard error 0.297, $R^2 = 0.272$)
  - One standard deviation more (12.9%, Table 1): price rise larger by 24.9 percentage points
  - Price bust -0.571 (0.083): 7.4 points deeper
  - Interpretation: more speculation, larger booms and larger busts

Back

Case Study, Complete Results (2/2)

- Table 2, Panels B–C (quantities in % of 2003 total volume; Section 3.2)
  - Unsold listings in the quiet: 5.961 (1.353), 76.9 points more per standard deviation
  - Foreclosures: 0.895 (0.398), 11.5 points, about 370 thousand more across the 115 areas
  - Interpretation: speculators who bought in the boom list in the quiet and fail to sell
  - That this turns the cycle is a model result (Sections 5–6), not identified by the regressions
- Figure 7, Panel C: a 10% price gain over the past year goes with an 8.2 points larger short-term buyer share
  - Interpretation: extrapolation draws short-horizon buyers in
- Table 6 (model): a 5% purchase tax
  - On all buyers: the price bust falls from 8.2% to 1.1%; on non-occupants only: to 5.8%
  - Interpretation: short horizons, not non-occupancy, amplify the cycle

Back

Case Study, Lead–Lag Inference

- p and v demeaned over the full sample; $k = -12, \ldots, 48$ months
  - Implied correlation $\hat{\beta}_k \, \mathrm{sd}(v_{t-k})/\mathrm{sd}(p_t)$
- At $k = 0$: 0.03; at $k = -12$: -0.34
  - Level correlations, not a directional test of whether prices lead volume
- Eq. (1) has no constant and the lag-aligned series need not have zero mean
  - The "implied correlation" is a standardised slope, not a Pearson correlation: bootstrap limits above 1 are possible
- The block bootstrap assumes stable, weakly dependent series
  - Doubtful for one boom–bust cycle in levels: the band is pointwise and exploratory
- 1990–2011: the maximum moves to the edge of the grid, $k = 48$ (0.82)

Back

The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: map what is known, then verify every reference
  - "Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design."
  - Resolve each DOI on Crossref and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses**: competing explanations, each with its own observable
  - "For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts."
  - E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code**: on the course data, replicating a known number first
  - "Implement GSADF with r0 = 0.01 + 1.8/sqrt(T) on weekly Bitcoin closes from the course data and simulate its 95% critical value."
  - Replicate first: GSADF = 3.85 and critical value 2.27 on 626 weeks, as in this chapter

Back

The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the alarms are evaluated
  - "`Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.`"
  - Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
  - "`Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?`"
  - Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid p-values, all 12 cells, the alarms that failed
  - The prompts that shaped the design go into `AI_USE.md`

Back

What the Researcher Must Check

- **Full-sample information in date-stamping**
  - Critical values, lag order or detrending fitted on the whole sample
  - Test: cut the sample at t and recompute; the flag at t must not change
- **Wrong tail**
  - Left-tail ADF critical values (about -2.86) used against an explosive alternative
  - Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility**
  - Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity**: many dates, assets and windows
  - Test: run the rule on simulated random walks and count paths with at least one alarm
- **Overlapping outcomes**: Fisher's exact or χ^2 tests overstate significance
  - Test: block bootstrap or circular rotations of the alarm series; drop windows cut by the end of the data

Back

Mini-Case: What the Result Does and Does Not Say

- A large one-sided p is not evidence of **no** information
  - It only fails to support a **higher** crash probability after alarms
  - Both point estimates are **negative**: one-sided rotation p for $\Delta < 0$, a lower crash rate after alarms, 0.07 (Bitcoin), 0.03 (Nasdaq 100)
  - Bitcoin: this 0.07 and the Newey–West one-sided 0.024 answer the same question under different null assumptions (stationary rotations vs a regression with HAC (Heteroskedasticity and Autocorrelation Consistent) errors)
  - Post-hoc, two-sided reading: alarms come in steady rises that did not end in a crash within 90 days
- Assumption behind the rotation p-values
  - Under H_0 every circular shift of the alarm series (at least 26 steps, ends joined) is as likely as the observed alignment
  - Stronger than equal hit rates: it needs stationary series and alarms independent of the crash process
  - Hence exploratory p-values; check their size by simulation under plausible null processes
- Block-bootstrap CI and Newey–West test of the same hit rates: Seminar 17, B7

Back

Further Study on Quantinar

Advanced courses on quantinar.com

- Cryptocurrency: Speculative Asset and Medium of Exchange
- Are Cryptos becoming alternative Assets?
- Statistics of Financial Markets

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

Back

Acronyms used in this material

- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **BET**: Bucharest Exchange Trading (index)
- **BET-FI**: BET Financial Investment companies index
- **BSADF**: Backward Sup Augmented Dickey–Fuller (statistic)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CI**: Confidence Interval
- **EM**: Expectation-Maximisation (algorithm)
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GSADF**: Generalised Sup Augmented Dickey–Fuller (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **JLS**: Johansen–Ledoit–Sornette (model)
- **LPPL**: Log-Periodic Power Law
- **LPPLS**: Log-Periodic Power Law Singularity (model)
- **LR**: Likelihood Ratio (test)
- **MC**: Monte Carlo (simulation)
- **NVIDIA**: NVIDIA Corporation (company name, US chip designer)
- **OLS**: Ordinary Least Squares
- **PSY**: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- **PWY**: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- **SADF**: Sup Augmented Dickey–Fuller (test)
- **SE**: Standard Error
- **SRISK**: Systemic RISK (capital shortfall measure)
- **US**: United States

References (1/4)

- Abreu, D., Brunnermeier, M.K. (2003). Bubbles and crashes. *Econometrica*, 71(1), 173–204.
- Astill, S., Harvey, D.I., Leybourne, S.J., Sollis, R., Taylor, A.M.R. (2018). Real-time monitoring for explosive financial bubbles. *Journal of Time Series Analysis*, 39(6), 863–891.
- Blanchard, O.J., Watson, M.W. (1982). Bubbles, rational expectations and financial markets. NBER Working Paper 945.
- Brunnermeier, M.K., Nagel, S. (2004). Hedge funds and the technology bubble. *Journal of Finance*, 59(5), 2013–2040.
- Carrasco, M., Hu, L., Ploberger, W. (2014). Optimal test for Markov switching parameters. *Econometrica*, 82(2), 765–784.
- Cheah, E.-T., Fry, J. (2015). Speculative bubbles in Bitcoin markets? An empirical investigation into the fundamental value of Bitcoin. *Economics Letters*, 130, 32–36.
- Conda, A., Ginavar, A., Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Cryptocurrency market analysis: insights from Metcalfe's law and log-periodic power laws. In *Handbook of Blockchain Analytics*, Springer, 63–79.
- DeFusco, A.A., Nathanson, C.G., Zwick, E. (2022). Speculative dynamics of prices and volume. *Journal of Financial Economics*, 146(1), 205–229.
- Demirer, R., Demos, G., Gupta, R., Sornette, D. (2019). On the predictability of stock market bubbles: evidence from LPPLS confidence multi-scale indicators. *Quantitative Finance*, 19(5), 843–858.
- Demos, G., Sornette, D. (2017). Birth or burst of financial bubbles: which one is easier to diagnose?. *Quantitative Finance*, 17(5), 657–675.
- Diba, B.T., Grossman, H.I. (1988). Explosive rational bubbles in stock prices?. *American Economic Review*, 78(3), 520–530.
- Evans, G.W. (1991). Pitfalls in testing for explosive bubbles in asset prices. *American Economic Review*, 81(4), 922–930.
- Fama, E.F. (2014). Two pillars of asset pricing. *American Economic Review*, 104(6), 1467–1485.

References (2/4)

- Filimonov, V., Sornette, D. (2013). A stable and robust calibration scheme of the log-periodic power law model. *Physica A*, 392(17), 3698–3707.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Garber, P.M. (1990). Famous first bubbles. *Journal of Economic Perspectives*, 4(2), 35–54.
- Geraskin, P., Fantazzini, D. (2013). Everything you always wanted to know about log-periodic power laws for bubble modeling but were afraid to ask. *European Journal of Finance*, 19(5), 366–391.
- Greenwood, R., Shleifer, A., You, Y. (2019). Bubbles for Fama. *Journal of Financial Economics*, 131(1), 20–43.
- Griffin, J.M., Shams, A. (2020). Is Bitcoin really untethered?. *Journal of Finance*, 75(4), 1913–1964.
- Hamilton, J.D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Hamilton, J.D. (1990). Analysis of time series subject to changes in regime. *Journal of Econometrics*, 45(1–2), 39–70.
- Hansen, B.E. (1992). The likelihood ratio test under nonstandard conditions: testing the Markov switching model of GNP. *Journal of Applied Econometrics*, 7(S1), S61–S82.
- Harvey, D.I., Leybourne, S.J., Sollis, R. (2015). Recursive right-tailed unit root tests for an explosive asset price bubble. *Journal of Financial Econometrics*, 13(1), 166–187.
- Harvey, D.I., Leybourne, S.J., Sollis, R., Taylor, A.M.R. (2016). Tests for explosive financial bubbles in the presence of non-stationary volatility. *Journal of Empirical Finance*, 38, 548–574.
- Homm, U., Breitung, J. (2012). Testing for speculative bubbles in stock markets: a comparison of alternative methods. *Journal of Financial Econometrics*, 10(1), 198–231.
- Johansen, A., Ledoit, O., Sornette, D. (2000). Crashes as critical points. *International Journal of Theoretical and Applied Finance*, 3(2), 219–255.

References (3/4)

- Kim, C.-J. (1994). Dynamic linear models with Markov-switching. *Journal of Econometrics*, 60(1–2), 1–22.
- Kindleberger, C.P., Aliber, R.Z. (2005). Manias, Panics and Crashes: A History of Financial Crises, 5th ed. Palgrave Macmillan.
- Pástor, Ľ., Veronesi, P. (2009). Technological revolutions and stock prices. *American Economic Review*, 99(4), 1451–1483.
- Pele, D.T. (2025). Dogecoin and speculative bubbles: a study using LPPL and sentiment analysis. Quantinar course.
- Pele, D.T., Mazurencu, M., Nijkamp, P. (2013). Herding behaviour, bubbles and log periodic power laws in illiquid stock markets: a case study on the Bucharest Stock Exchange. SSRN Working Paper.
- Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Metcalfe's law and log-period power laws in the cryptocurrencies market. *Economics*, 13(1), 2019-29.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- Phillips, P.C.B., Magdalinos, T. (2007). Limit theory for moderate deviations from a unit root. *Journal of Econometrics*, 136(1), 115–130.
- Phillips, P.C.B., Shi, S. (2018). Financial bubble implosion and reverse regression. *Econometric Theory*, 34(4), 705–753.
- Phillips, P.C.B., Shi, S. (2020). Real time monitoring of asset markets: bubbles and crises. In *Handbook of Statistics: Financial, Macro and Micro Econometrics Using R*, Elsevier, 61–80.
- Phillips, P.C.B., Shi, S., Yu, J. (2014). Specification sensitivity in right-tailed unit root testing for explosive behaviour. *Oxford Bulletin of Economics and Statistics*, 76(3), 315–333.
- Phillips, P.C.B., Shi, S., Yu, J. (2015a). Testing for multiple bubbles: historical episodes of exuberance and collapse in the S&P 500. *International Economic Review*, 56(4), 1043–1078.

References (4/4)

- Phillips, P.C.B., Shi, S., Yu, J. (2015b). Testing for multiple bubbles: limit theory of real-time detectors. *International Economic Review*, 56(4), 1079–1134.
- Phillips, P.C.B., Wu, Y., Yu, J. (2011). Explosive behavior in the 1990s Nasdaq: when did exuberance escalate asset values?. *International Economic Review*, 52(1), 201–226.
- Scheinkman, J.A., Xiong, W. (2003). Overconfidence and speculative bubbles. *Journal of Political Economy*, 111(6), 1183–1220.
- Shi, S., Phillips, P.C.B. (2023). Diagnosing housing fever with an econometric thermometer. *Journal of Economic Surveys*, 37(1), 159–186.
- Shiller, R.J. (2015). Irrational Exuberance, revised and expanded 3rd ed. Princeton University Press.
- Shu, M., Zhu, W. (2020). Detection of Chinese stock market bubbles with LPPLS confidence indicator. *Physica A*, 557, 124892.
- Sornette, D. (2017). Why Stock Markets Crash: Critical Events in Complex Financial Systems. Princeton University Press.
- Vasilopoulos, K., Pavlidis, E., Martínez-García, E. (2022). exuber: recursive right-tailed unit root testing with R. *Journal of Statistical Software*, 103(10).
- Whitehouse, E.J., Harvey, D.I., Leybourne, S.J. (2023). Real-time monitoring of bubbles and crashes. *Oxford Bulletin of Economics and Statistics*, 85(3), 482–513.
- Whitehouse, E.J., Harvey, D.I., Leybourne, S.J. (2025). Real-time monitoring procedures for early detection of bubbles. *International Journal of Forecasting*, 41(3), 1260–1277.